VISUAL DESIGNS

Greenfield Local Hub

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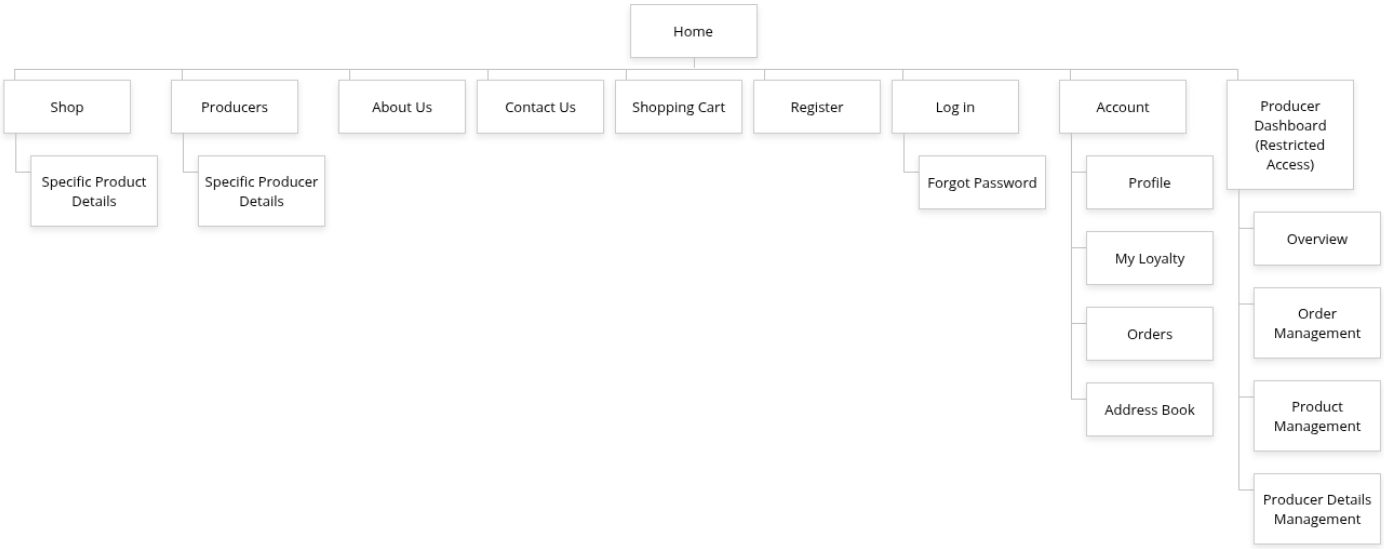
Site Map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Wireframe Designs

Global Style Guide

In this style guide I will go through the typical elements and key aspects of the wireframes such as the colour scheme, typography, layout and spacing, navigation bar and the footer. This style guide will document all recurring elements of the wireframes, removing the need to repeat this information in each page's individual annotations.

Typography

The fonts I have used across all the pages of the Greenfield Local Hubs wireframes are two, a sans-serif and a serif font. The serif font I used was "Playfair Display" and it is mainly used for all headings, hero titles, section titles and product prices. The font represents elegance and gives the user a feeling of sincerity but also makes the website have a vintage business aesthetic. Now for the sans-serif font I used for the wireframes is "DM sans", it is used for all other body text, navigation, footer, buttons footer, and form. It is a simple calm font that is easy to read and has a modern style to it. Both fonts are freely available in Canva and sourced from a trusted open-source which is google fonts. This is great since the source is dependable and could be confidently used when developing the solution.

The sizes I have used for the fonts are various, the hero titles are 60px on the wireframes. Section titles are 16px, and for the other body texts the sizes vary. The font sizes maintain visual balance creating a great user experience making it readable for the users.

Colour Scheme

Main colours used in the wireframes are dark green, a warm brown and white. The green and brown were used because of the natural landscape that represent the farmers and food producers. The dark green (#1B4D3F) was used consistently for primary buttons, the Greenfield Local Hub logo, and for key navigation elements including the footer. The brown (#7C5C3E) representing the earth, was used on badges, and few buttons which helps draw the viewers attention to read the element. The white used in the designs make sure of high contrast and readability, and it makes the design look more appealing to the eyes giving a sense of breathing space when navigating through the solution. It is used in the background of the designs, the navbar, and in forms. These colours together support accessibility and usability while also strengthening the brands theme.

Navigation & Footer

The white navigation bar is positioned at the top of every page with a slight drop shadow. The navbar design and positioning will stay consistent across every page without any exceptions. It consists of the business logo, the navigational links such as home, shop, producers, about us, and contact. It also has a shopping cart, account information, register and sign in. The elements inside are mostly grey and white, except for the register button and the logo being the dark green mentioned in the colour scheme. The navigation bar follows a common convention structure, having every necessary element that all professional digital solutions include. It is professional and slick and easy for the user to navigate with which is the main goal of a navigation bar. Just above the navbar is a green accessibility bar which will offer a drop-down menu of accessibility features when hovering on the buttons. There will be high contrast mode, large text, small text, dark mode, screen reader, etc.

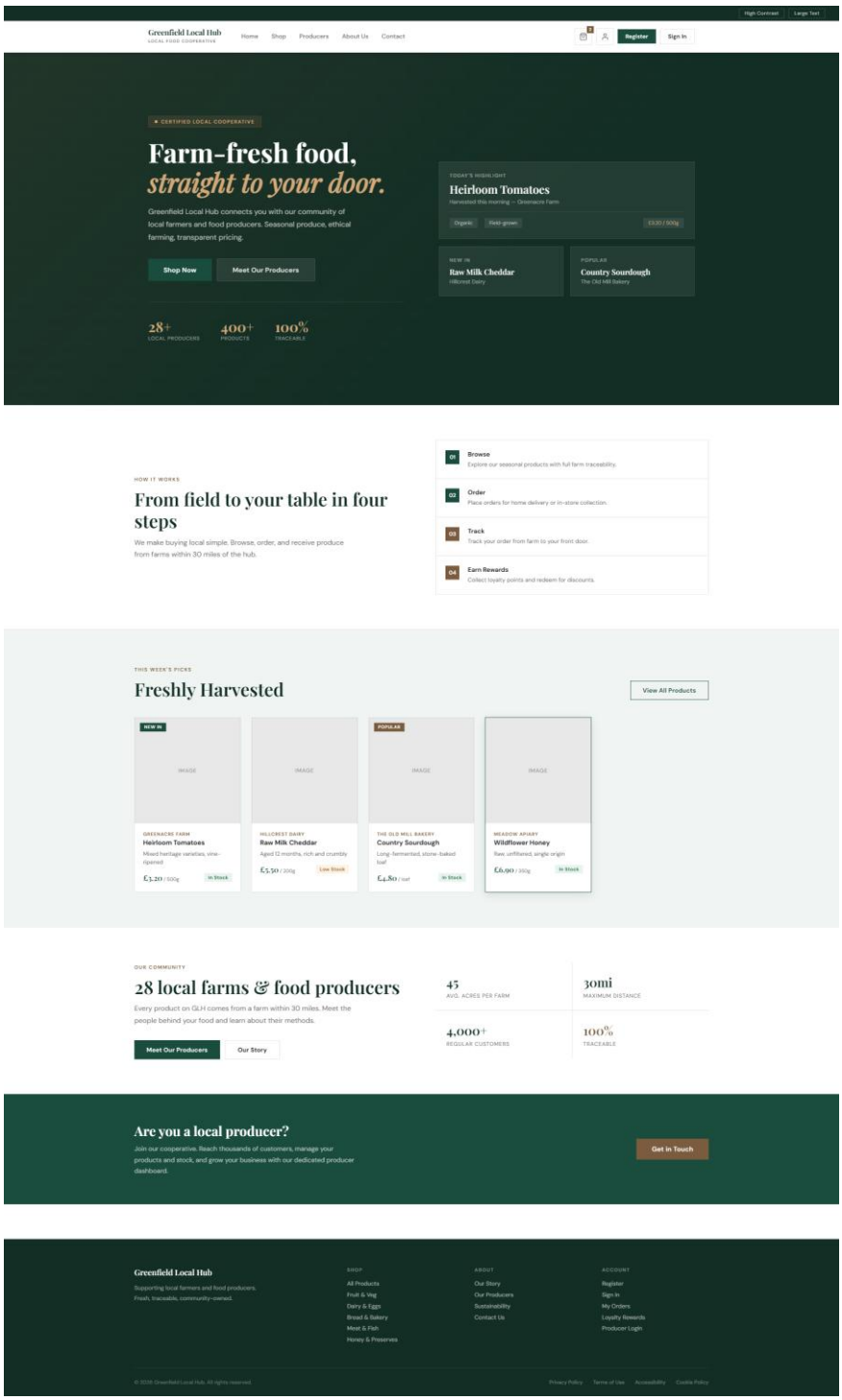
Similarly, the footer is consistent with its design and positioning across all the pages in my wireframes. With its bottom of the page positioning, which is industry standard, it consists of quick links to various pages and a small business identity section. The footers colour is the same dark green mentioned previously, and the quick links are in a shade of grey, and the business name is written in white. The quick links will eventually send the user to different pages of the solution, some will even take the user to a filtered page such as "Fruit & Veg". If the user clicks this link they will be redirected to the shop, but it will filter the products to only show the fruits and vegetables sold. These

quick links will be useful and convenient for the user, enhancing their user experience and engagement of the solution.

Layout

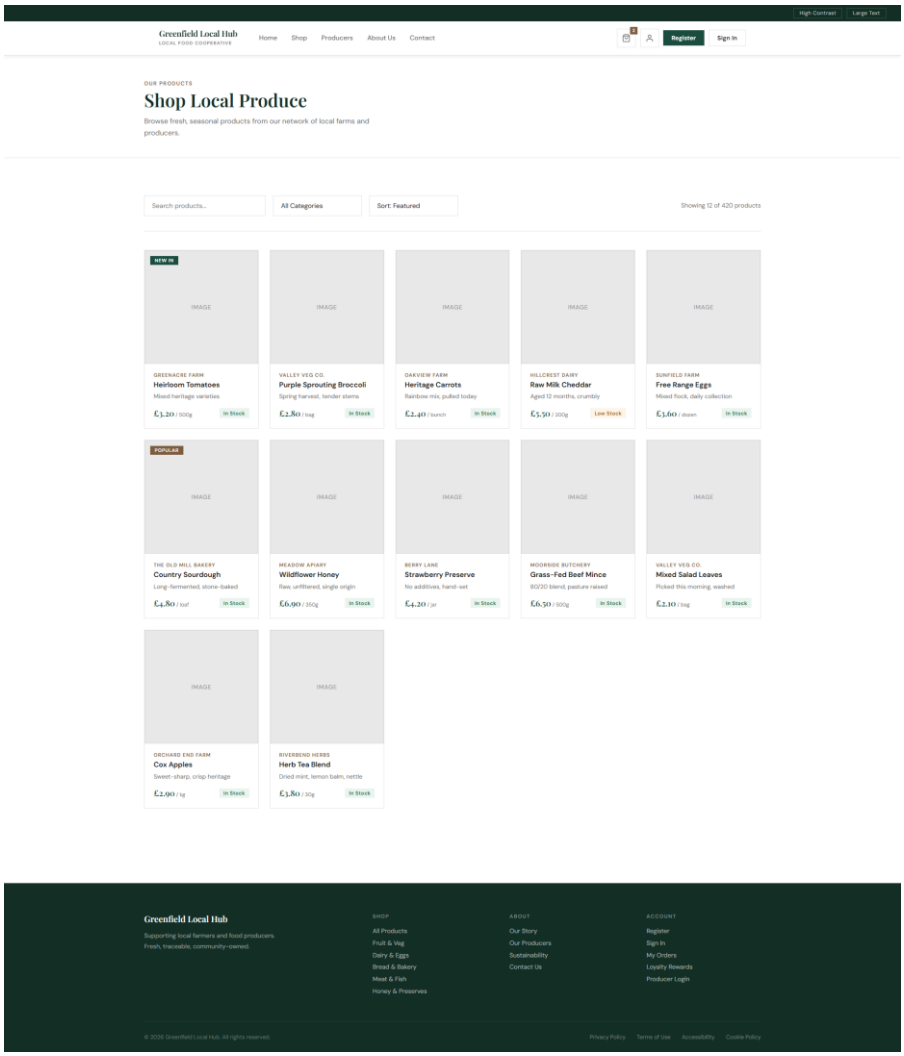
Almost every page I designed begins with a page hero section just underneath the navigation bar, and it includes descriptive text suitable for the selected page. The pages are designed in sections, every section has a different colour from the section after, this creates visual separation between the content. I have tried to maintain a consistent design through out all the pages in order to achieve a professional industry standard product.

Home Page



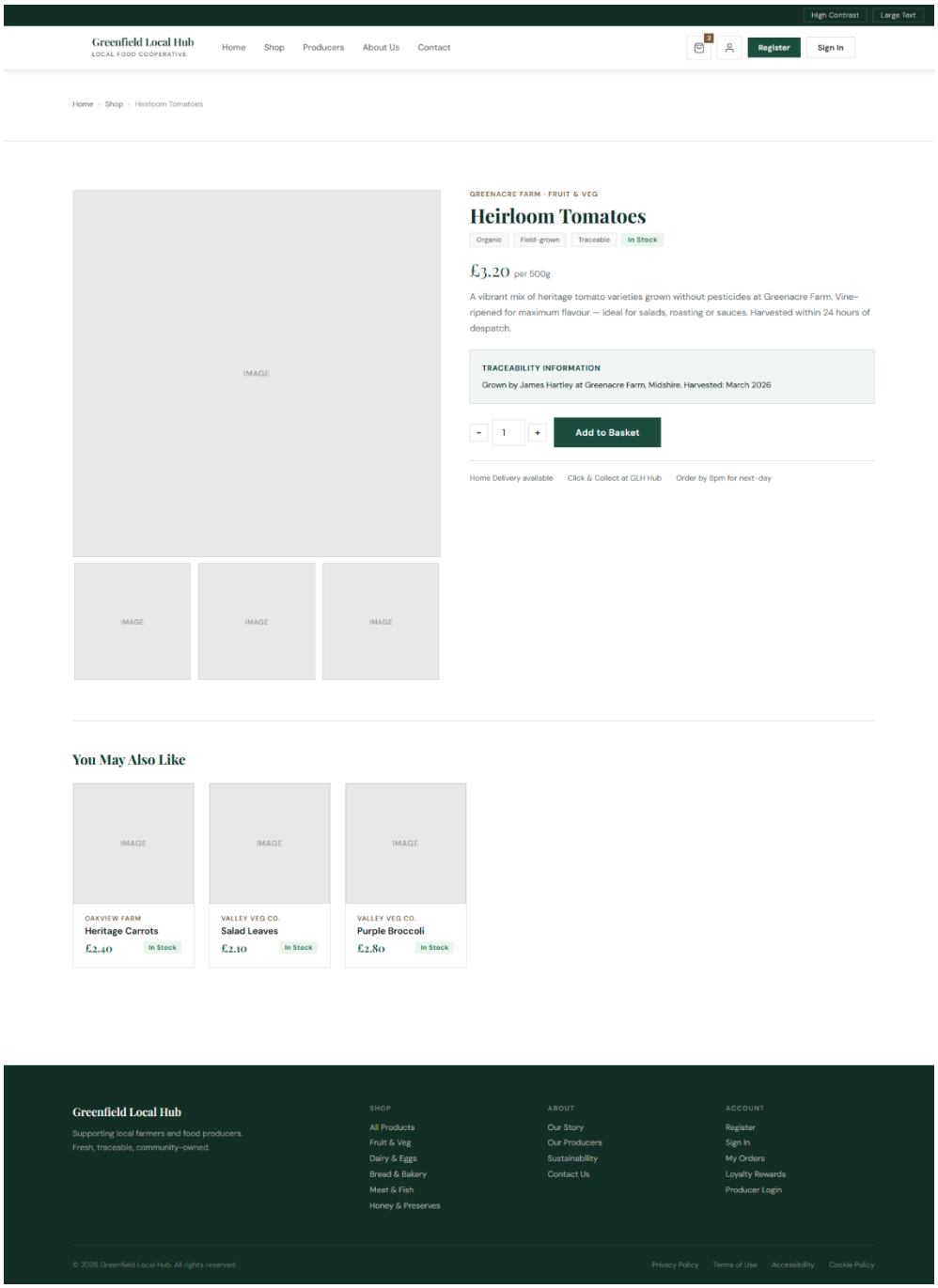
This is the design of the home page, and it presents Greenfield Local Hubs brand and value while leading users to key actions. All global elements such as the navigation bar and footer follow the descriptions mentioned in the Global Style Guide. The page starts with a hero section containing the business slogan and a quick description of the business, it also has a couple of important information that could encourage engagement such as today's highlight, new in and popular items. The next hero section displays the actions that the user could take such as browsing, ordering, tracking and earning points. Right after that popular items and the weeks picks are presented, these products will interest the user and may lead them to view the shop and make a purchase. The last two sections provide additional information and an action button for local producers who are interested in joining cooperative.

Shop Page



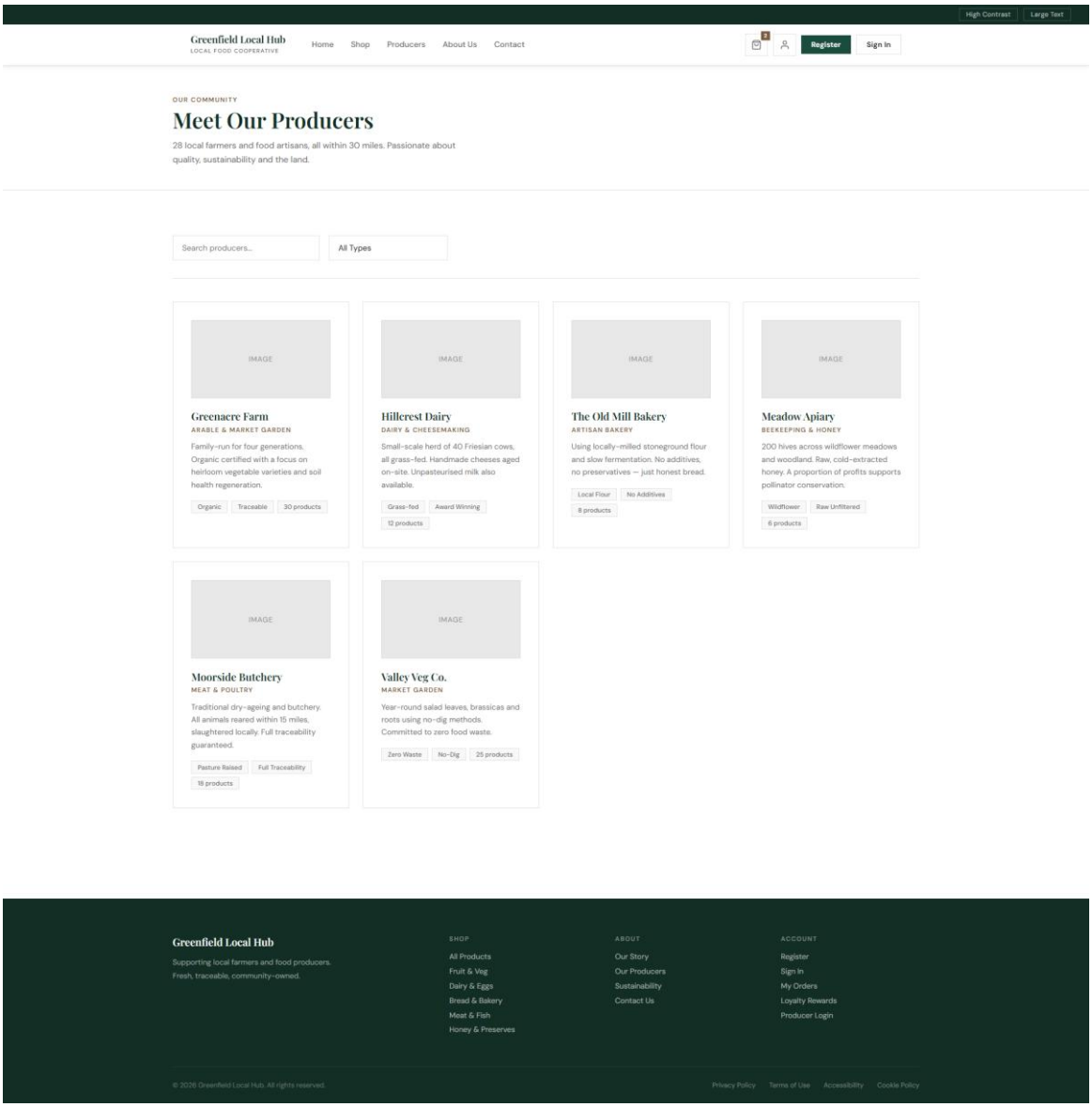
The shop displays all the products sold on the site, each product card shows the producer of the product, product name, price and the availability of the item. This meets the requirements that were mentioned by GLH, displaying transparency with the products. There is a search feature in the design, this will let users search for any specific product in the page. Next to the search there are filtering features, user will be able to filter by category, sort by popularity, sort by new items, sort by price from ascending or descending order. These features will help the user navigate and find their desired items efficiently.

Specific Product Page (Product: Heirloom Tomatoes)



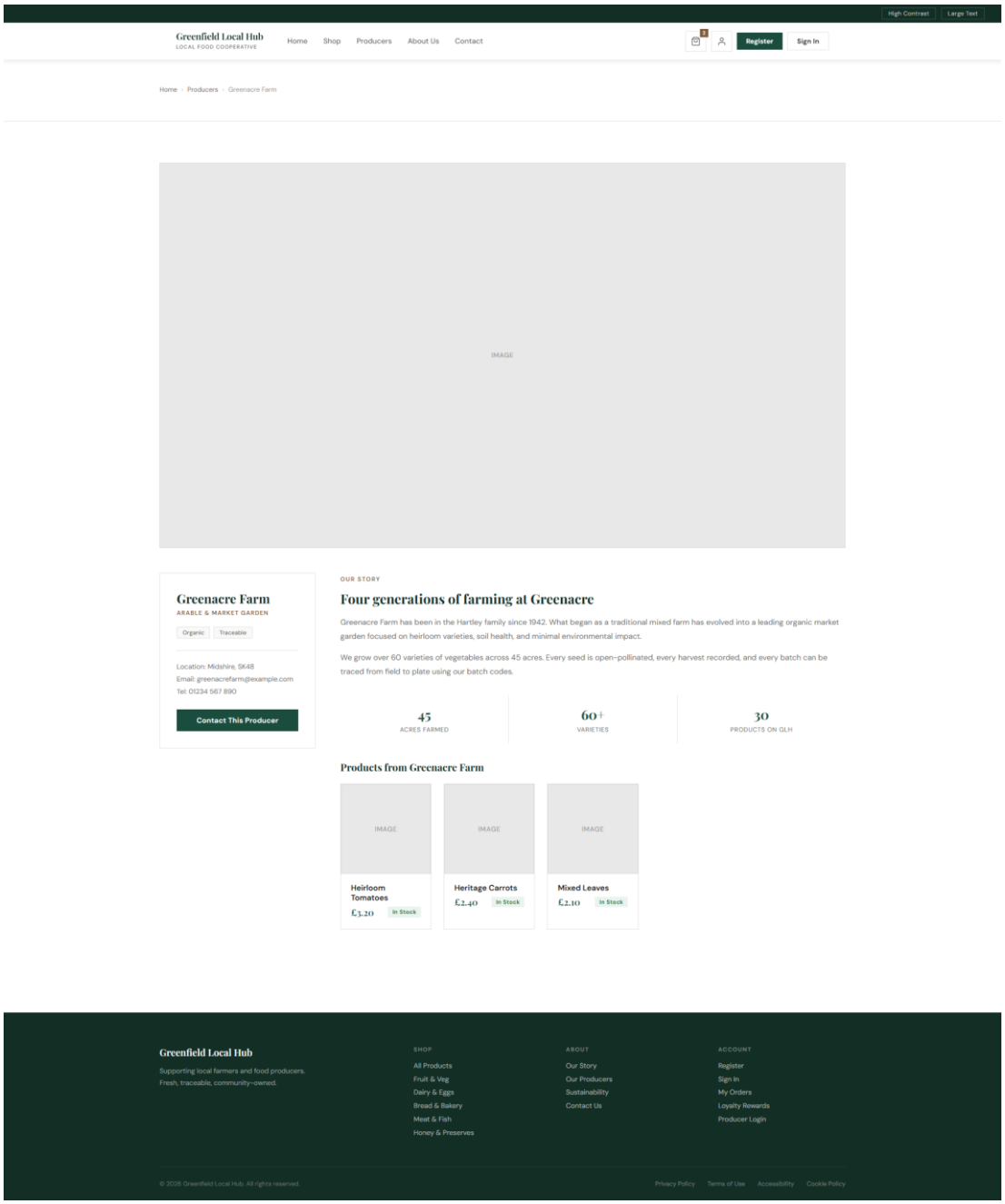
When the user clicks a specific product, they want to view or purchase, a page for that product will appear as you can see in the design. It will show four different images of the product, and it will show the producer, category of the item, price per unit, and a description of the product. There will also be traceability information displayed, and of course the add to basket button and the ability of choosing the quantity to add. A useful section was designed below the main item information, a “You May Also Like” section. The user will be able to look at other products that they may like based on the product they clicked. This feature is appreciated by users since it enhances their user experience, and it is a common convention in digital solutions and could encourage additional purchases.

Producers Page



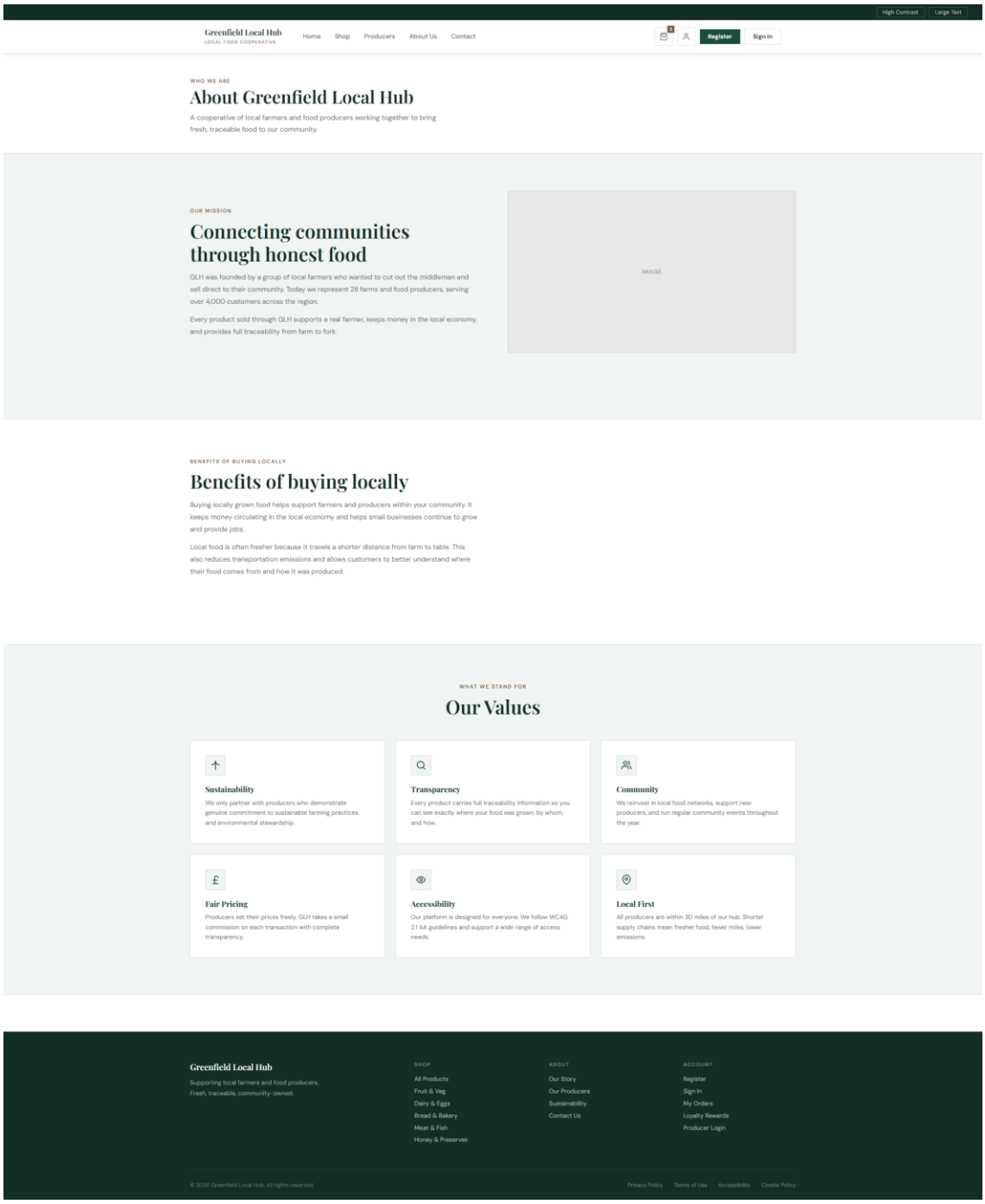
The producer page displays all the producers with images, what type of producer they are, and a brief description of each producer. Above the producer cards, there is a search bar as well making it easier for users to find a specific user. There is also sorting by type features, such as sorting producers for bakery, dairy, meat & poultry, etc. The design successfully includes necessary information that users are interested in, this will increase the businesses trust in the community.

Specific Producers Page (Producer: Greenacre Farm)



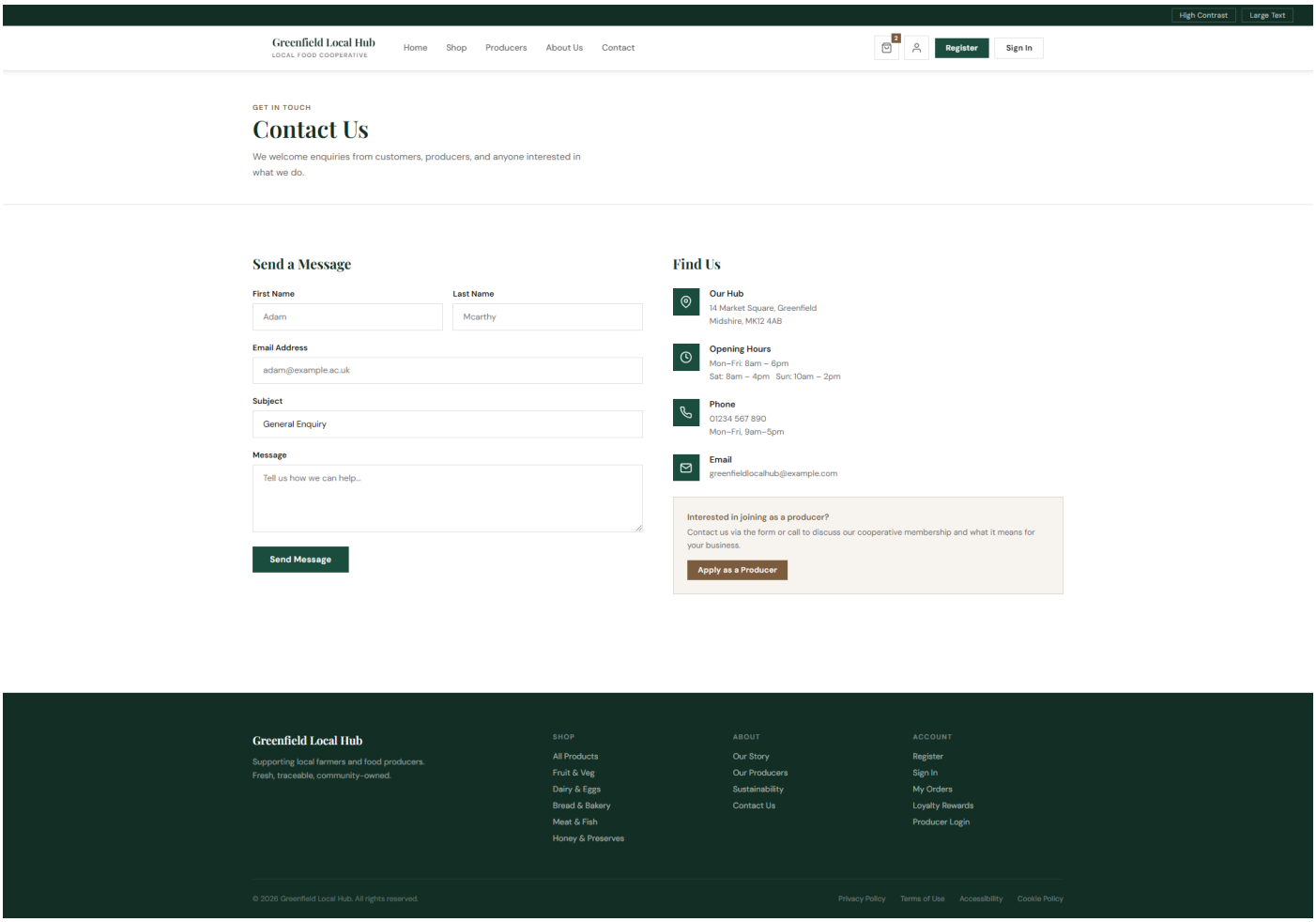
When the user clicks a specific producer, a dedicated page for that producer will appear. This page will include additional information about the producer, what methods they use to produce, and all their contact details and how to find them is available in the page as well. There is also a section below the main information that will display the products sold from that specific producer, this will help promote the producer and encouraging more sales.

About Us Page



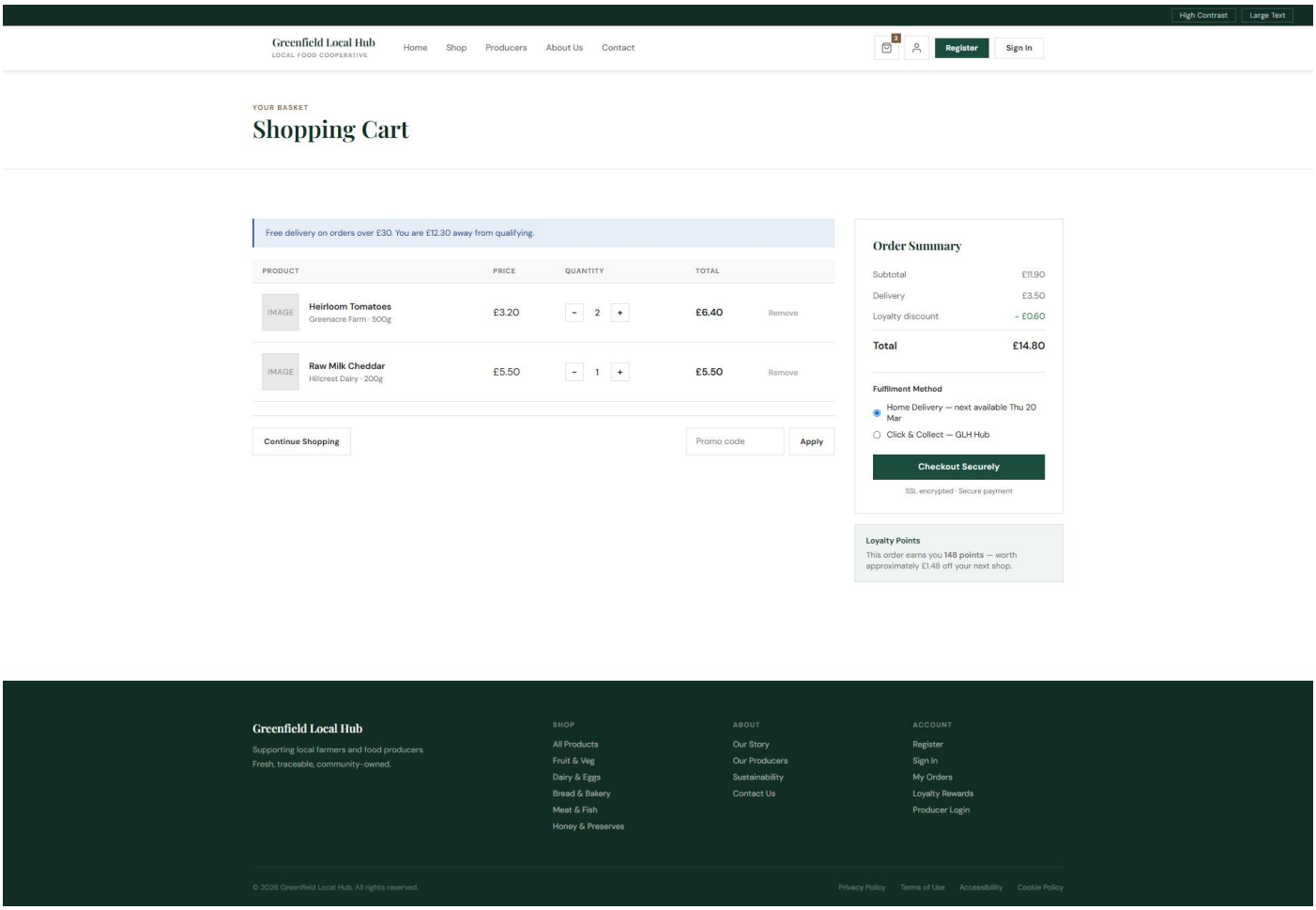
The about us page shows users who Greenfield Local Hub is and what they stand for. It starts with a short introduction explaining the business idea and what they do. Next section explains the mission of the business, and the one after about the benefits of buying locally. This was a requirement mentioned by the client to include in the digital solution. Then there is a values section, describing six principles that guide the business.

Contact Page



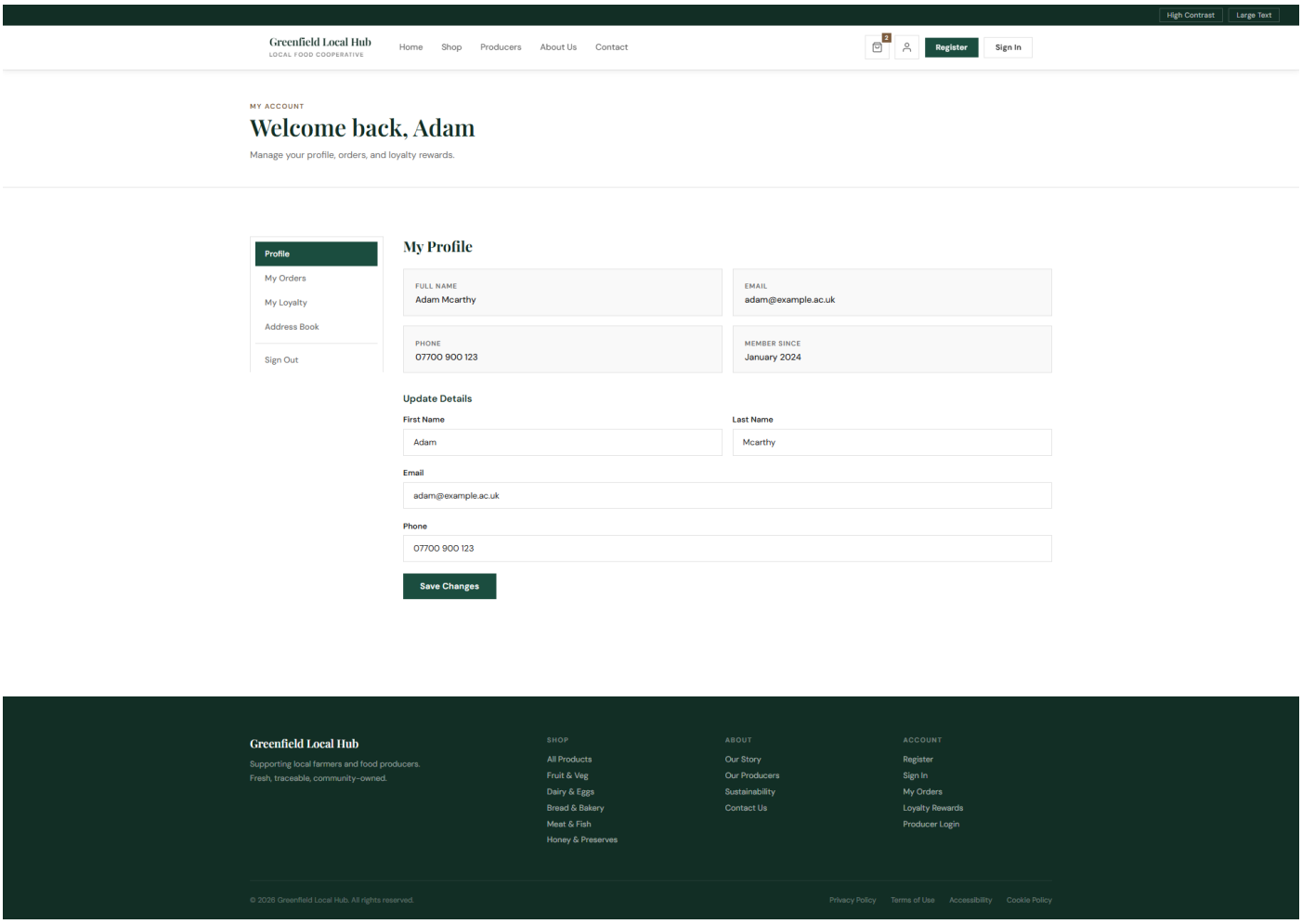
In this contact page the user will be able to send an enquiry to Greenfield Local Hub. It also clearly displays the location of the hub, opening hours, phone number and email address. If there are any local producers interested in joining the cooperative, they can also apply from this page using the brown action button.

Shopping Cart Page



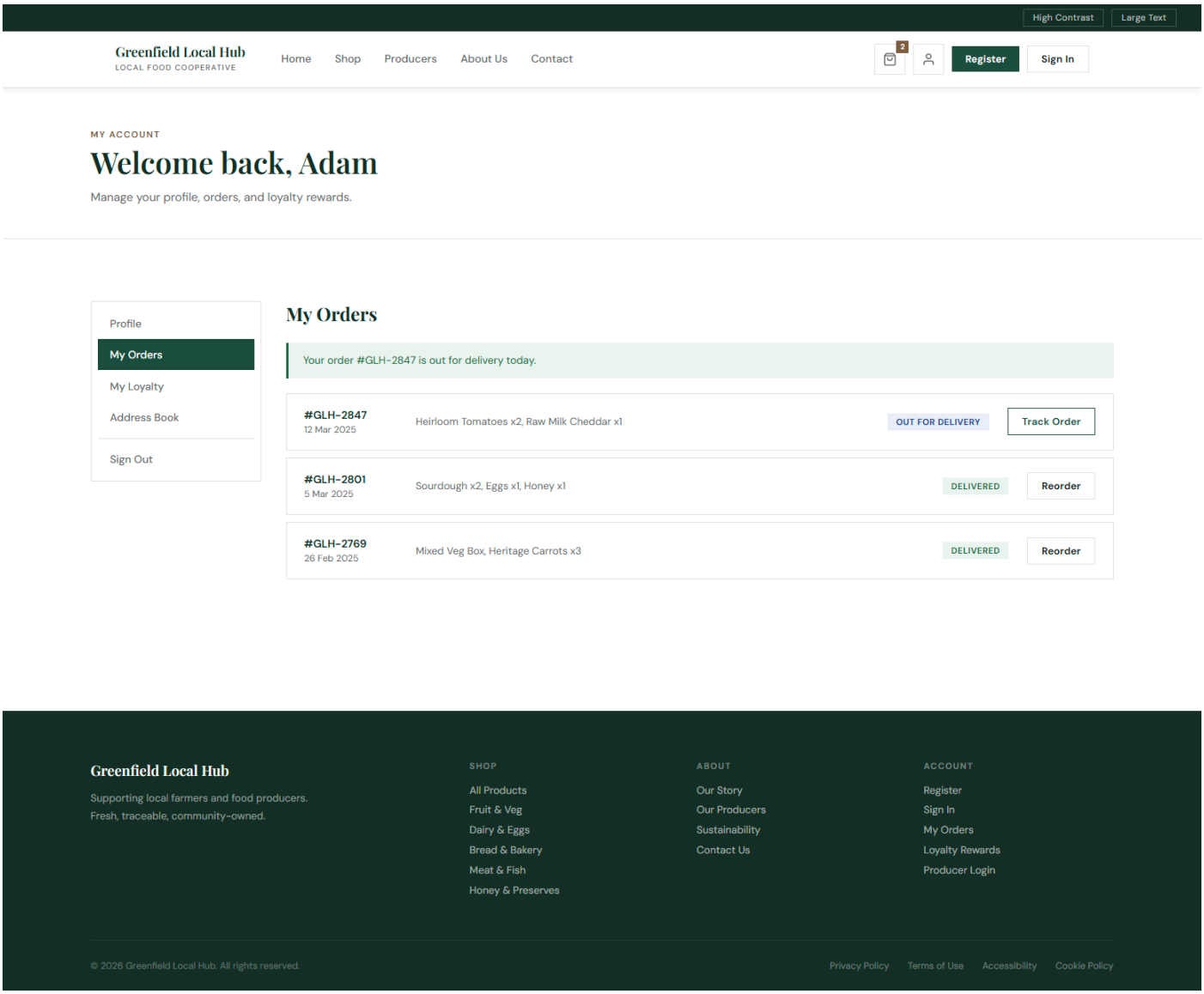
The shopping cart page will display all the items saved in the basket by the user. Each product shows its price, quantity and total price. Users will be able to remove product, increase/decrease quantity or apply promo code from the shopping cart. When the user wants to check out, they can easily see the total sum to pay, and they will have to choose to order for delivery or collection at the hub.

Account Profile Page



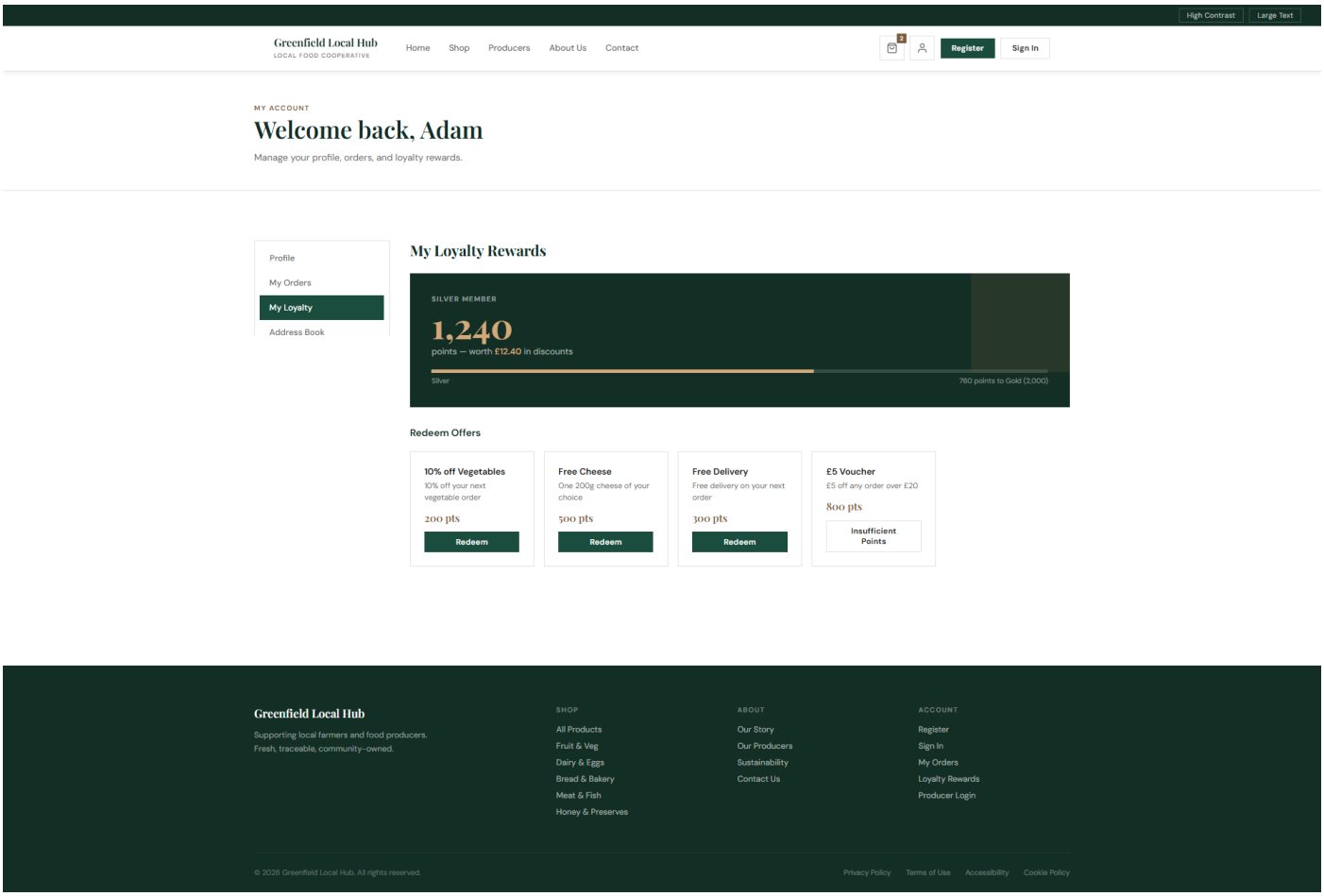
This is the account profile page which shows the logged in users personal information and that includes the name, email address, phone number and the time of joining the site. The user will be able to update their details if needed by simply filling the form and pressing save.

My Orders Page



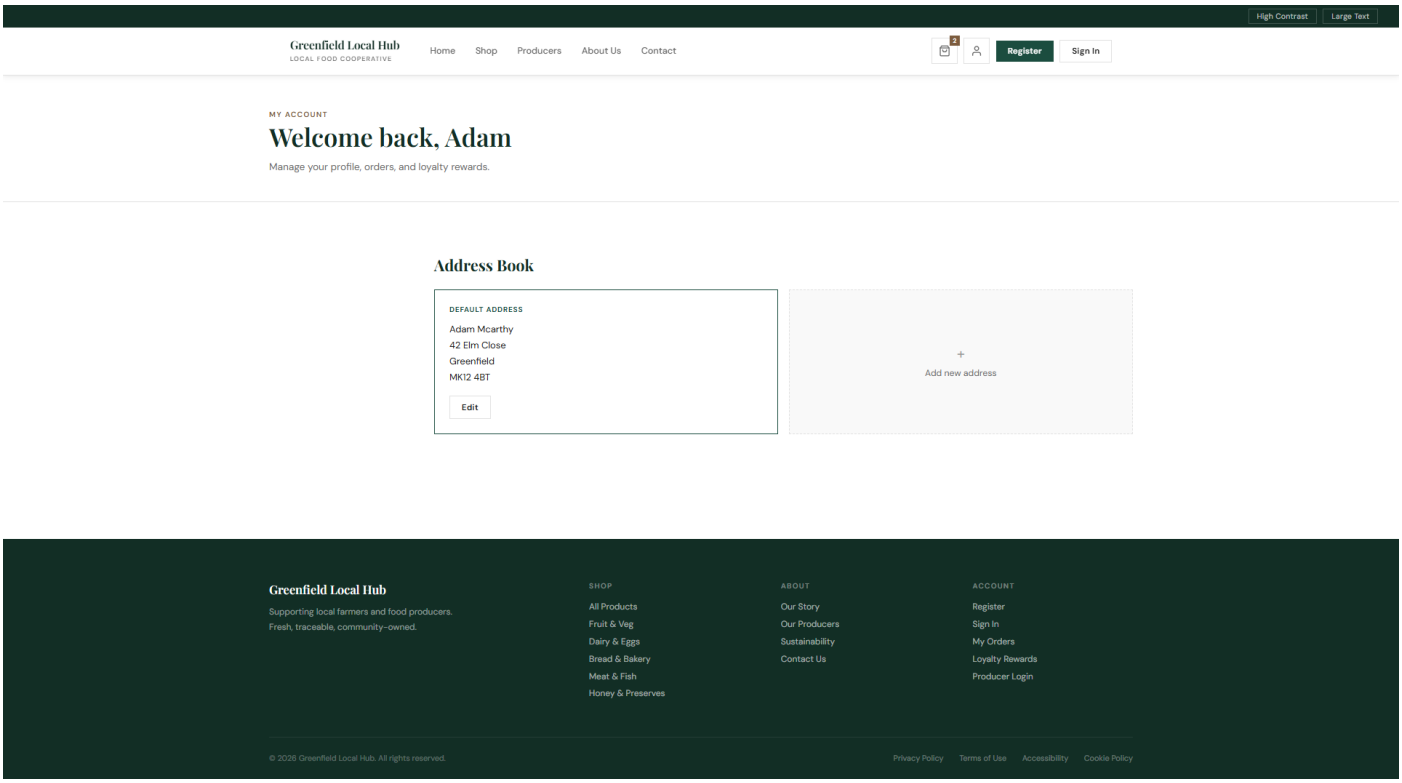
In the My Orders page, the user can view all previous orders and current orders. The user can see the order status and even able to track the order. The user can see what products were purchased in every order, this will make the staffs lives easier by decreasing enquiries from customers. The tracking feature was mentioned to be included by the client, and it will help users be more relieved knowing the status of their orders.

My Loyalty Page



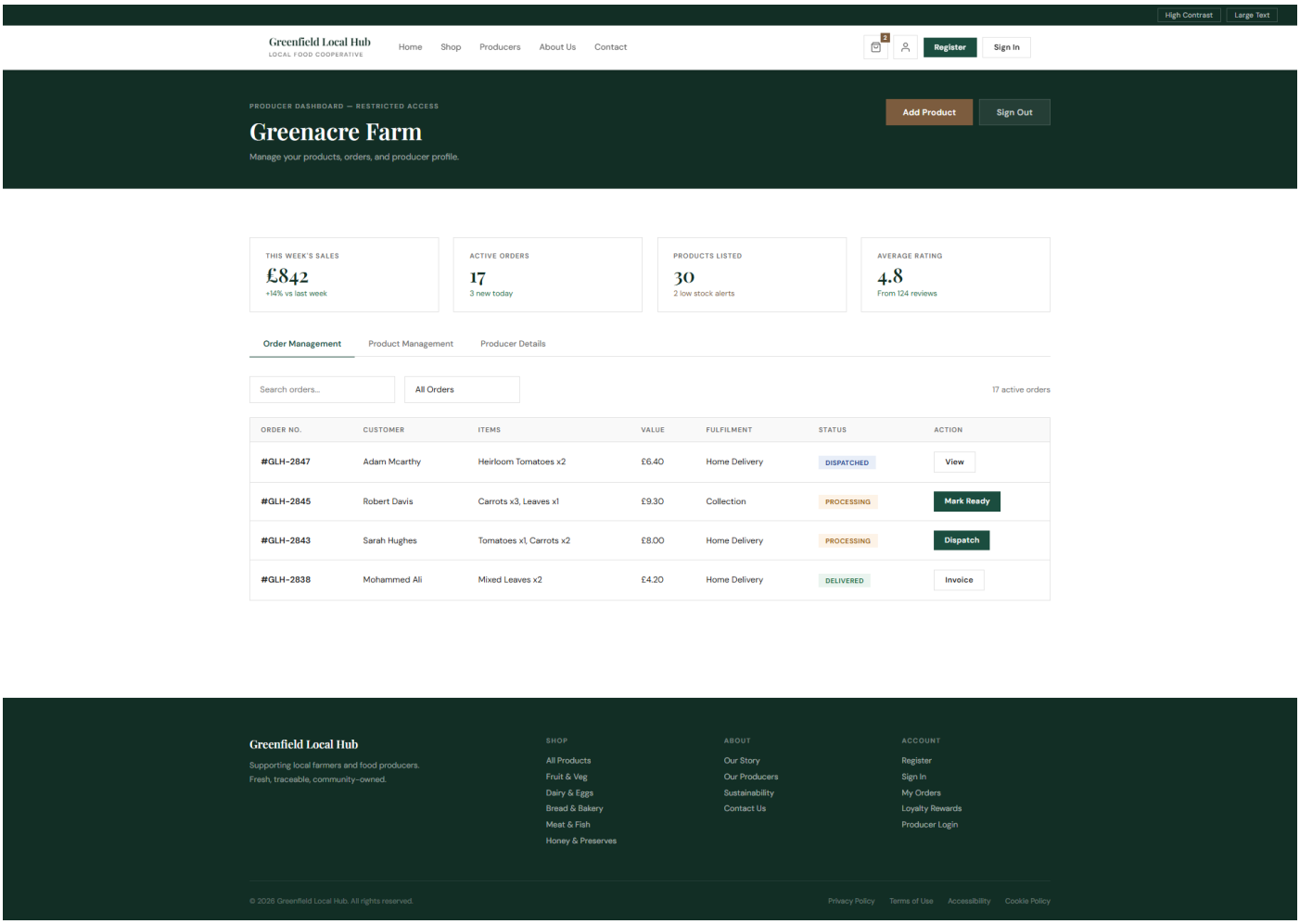
The loyalty page shows all the loyalty points the user has saved from all their orders. The points are given depending on the sum of each order, and the user will be able to redeem discount vouchers and products using their points.

Address Book Page



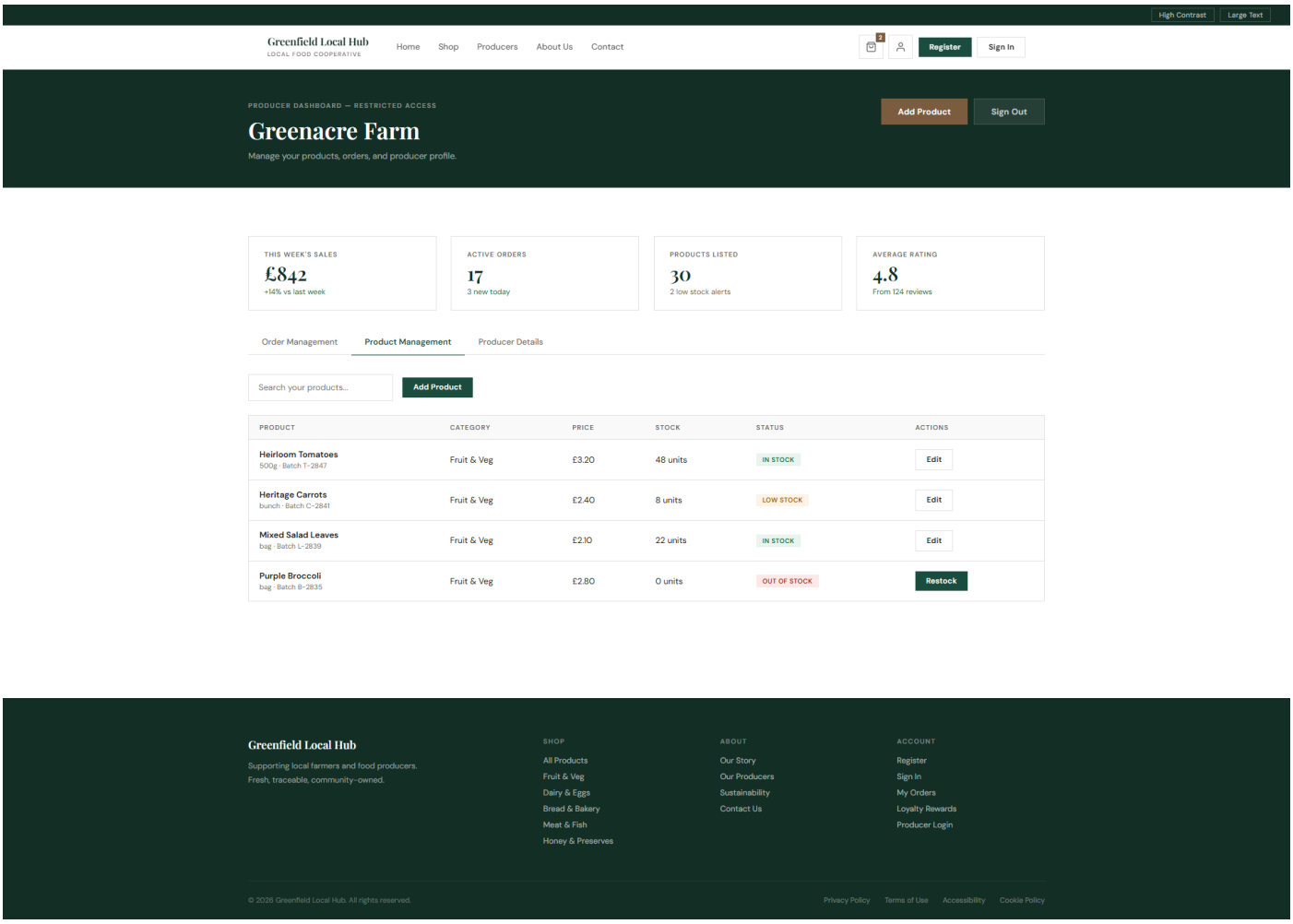
In order for users to place orders by delivery, they will need to have an address saved. Here in the address book the user can save multiple addresses for deliveries. The addresses could be easily edited, and new ones could also be added. This is convenient if a user wants to buy on behalf of someone else, so they could just add a new address quickly.

Producer Dashboard Page (Order Management Section)



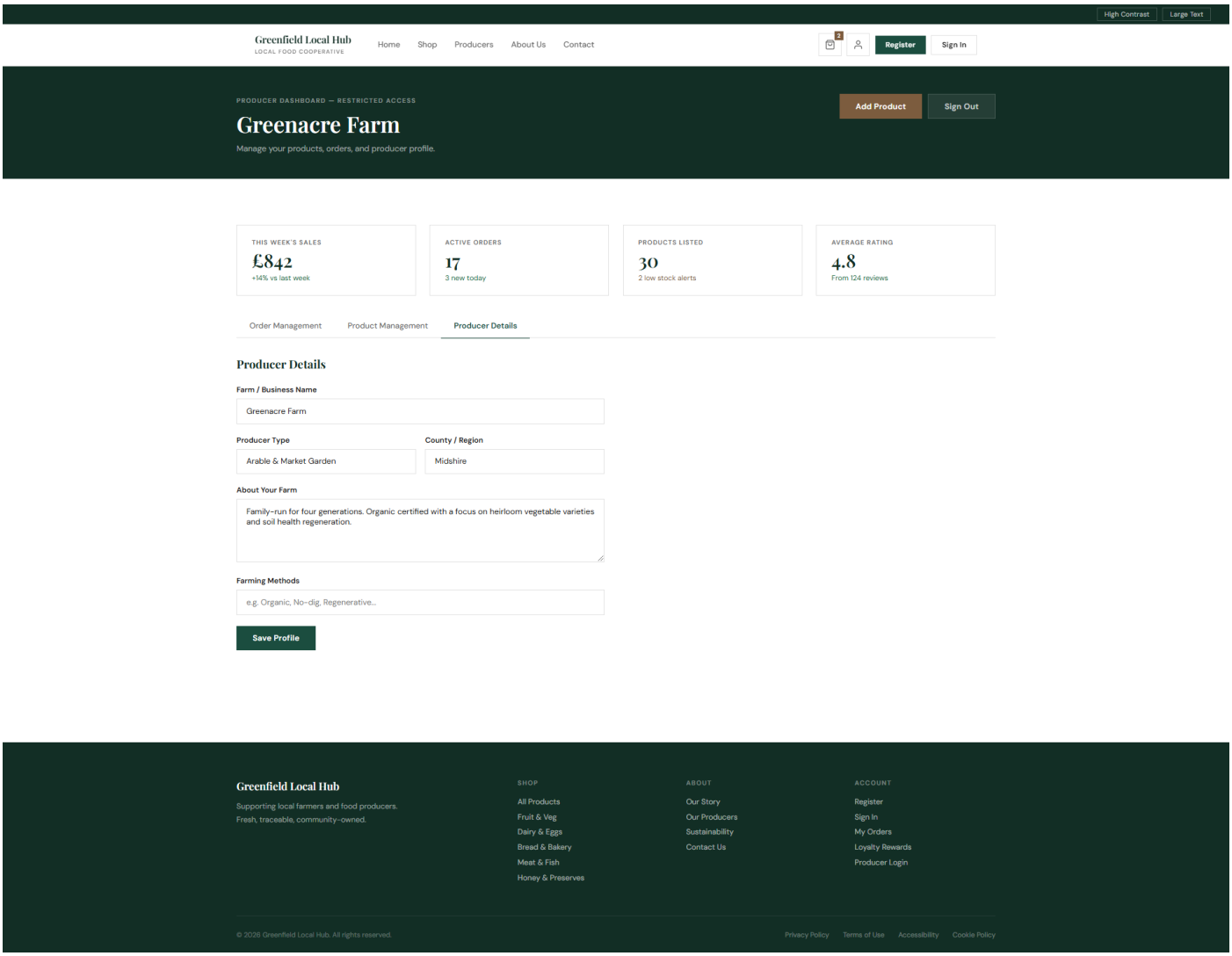
This is the producer dashboard accessed only by producers. This is the order management section specifically, and here can the user manage all the customer's orders. In small cards the producer can see an overview of their business statistics: this week's sales, active orders, product listed and average rating. Below the cards, a table displays each customer order with key details like the name of the customer, items bought, value of the order, if it is for delivery or collection and the status of the order. The producer is then able to change the action of the order, which will be shown to user when they track their order.

Producer Dashboard Page (Product Management Section)



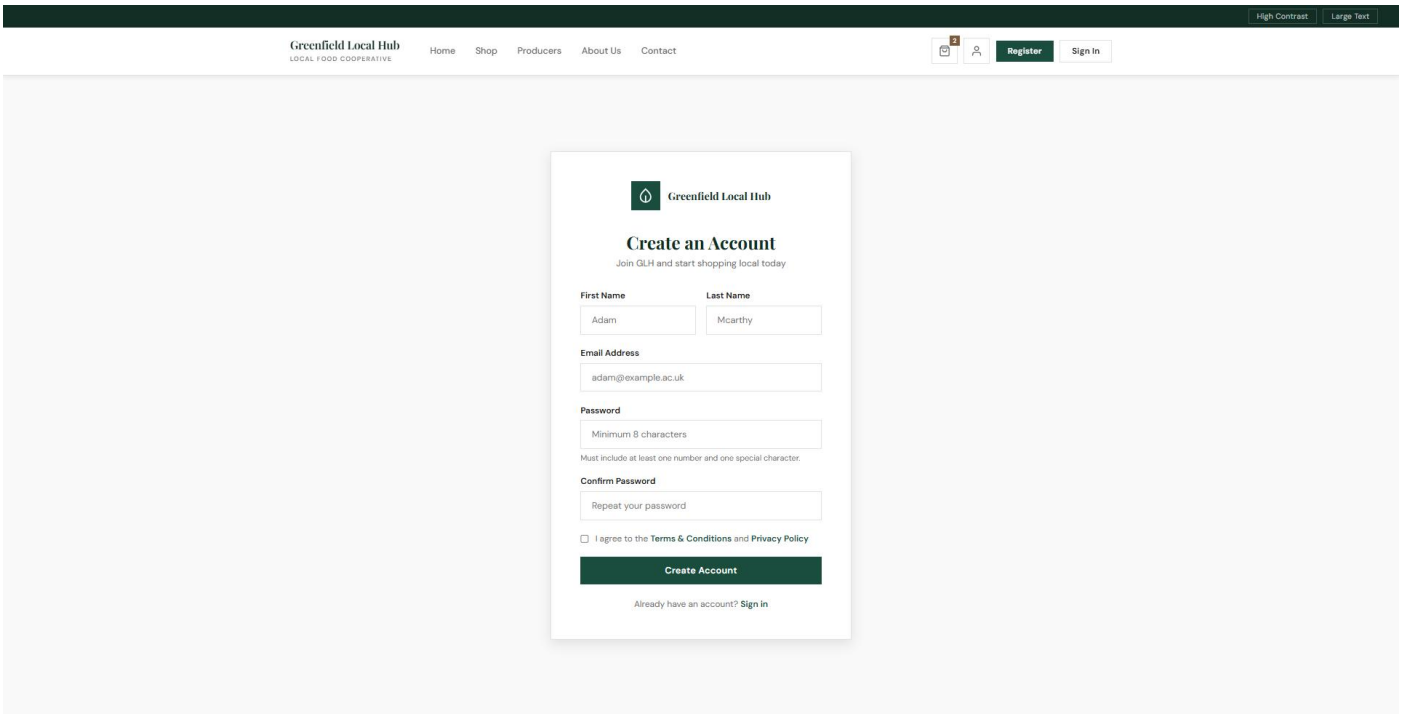
This is the product management section of the dashboard, the producer can add or edit existing products here. The producer could user the search bar to find a specific product to edit efficiently, below the search is a table containing all the products. It includes key information such as the category of each product, price, stock and status. If the producer would like to restock or change a products details or price, they could just go to edit of the specific product and make the changes.

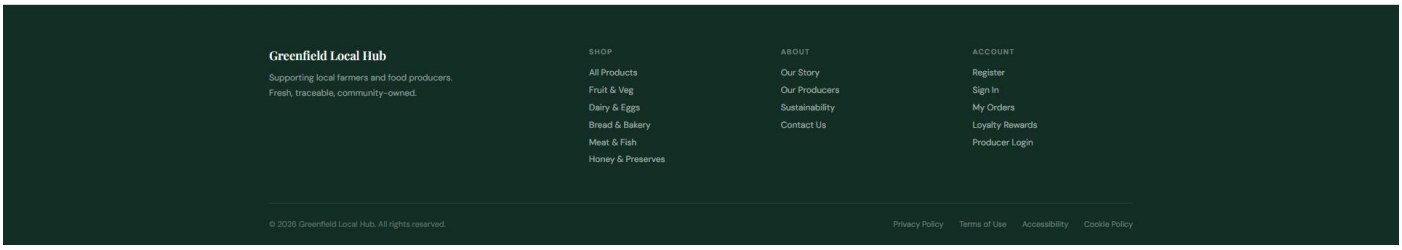
Producer Dashboard Page (Producer Details Section)



Last section of the producer dashboard is the producer details. In this page, the producer could update their details which are shown in the producer page users. By simply filling the form and saving, the producer details will update for the customers when viewing.

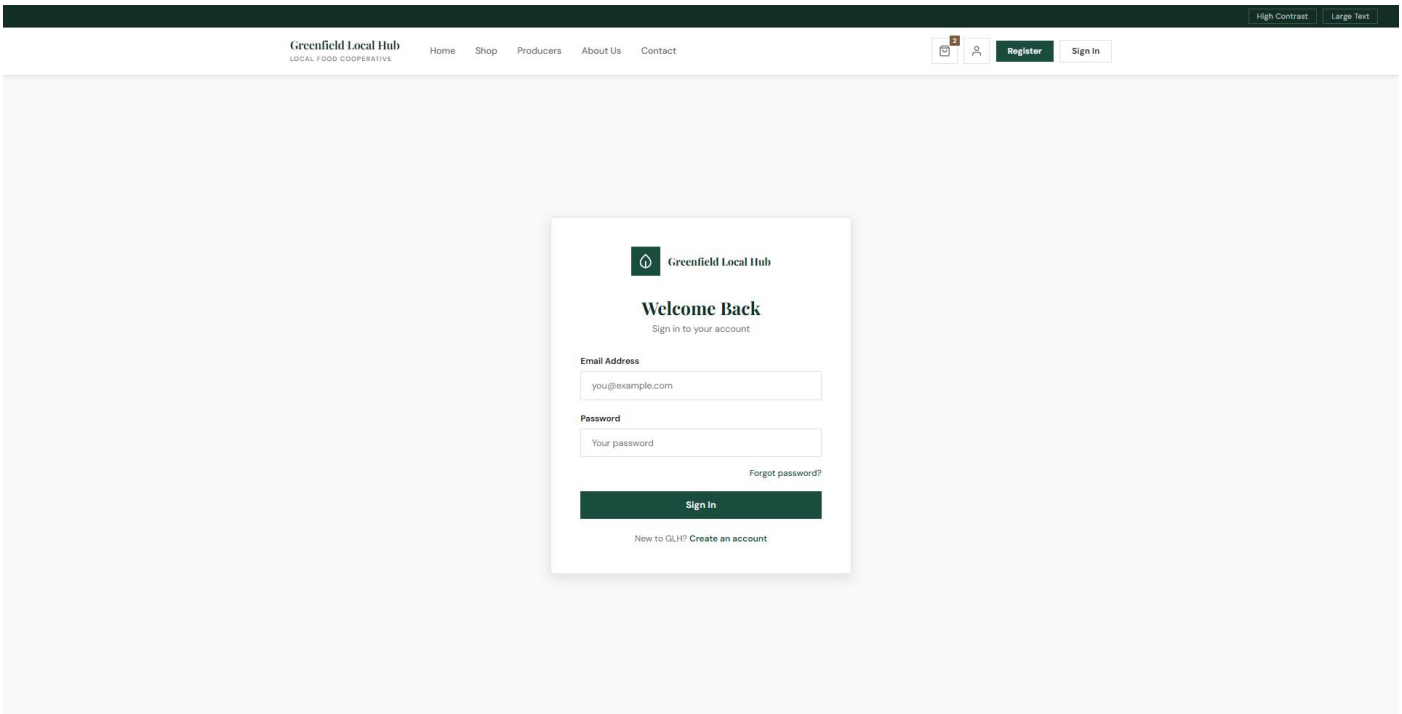
Register Page

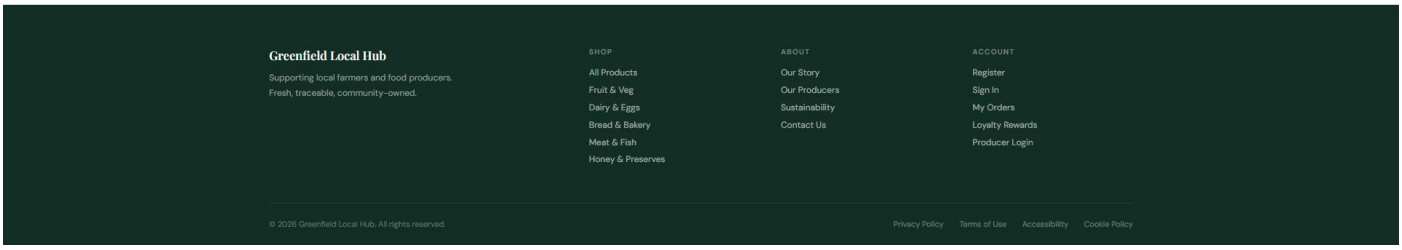




Here is the register page, designed in an industry standard way, it includes all necessary information and successfully follows the styling mention in the Global Style Guide.

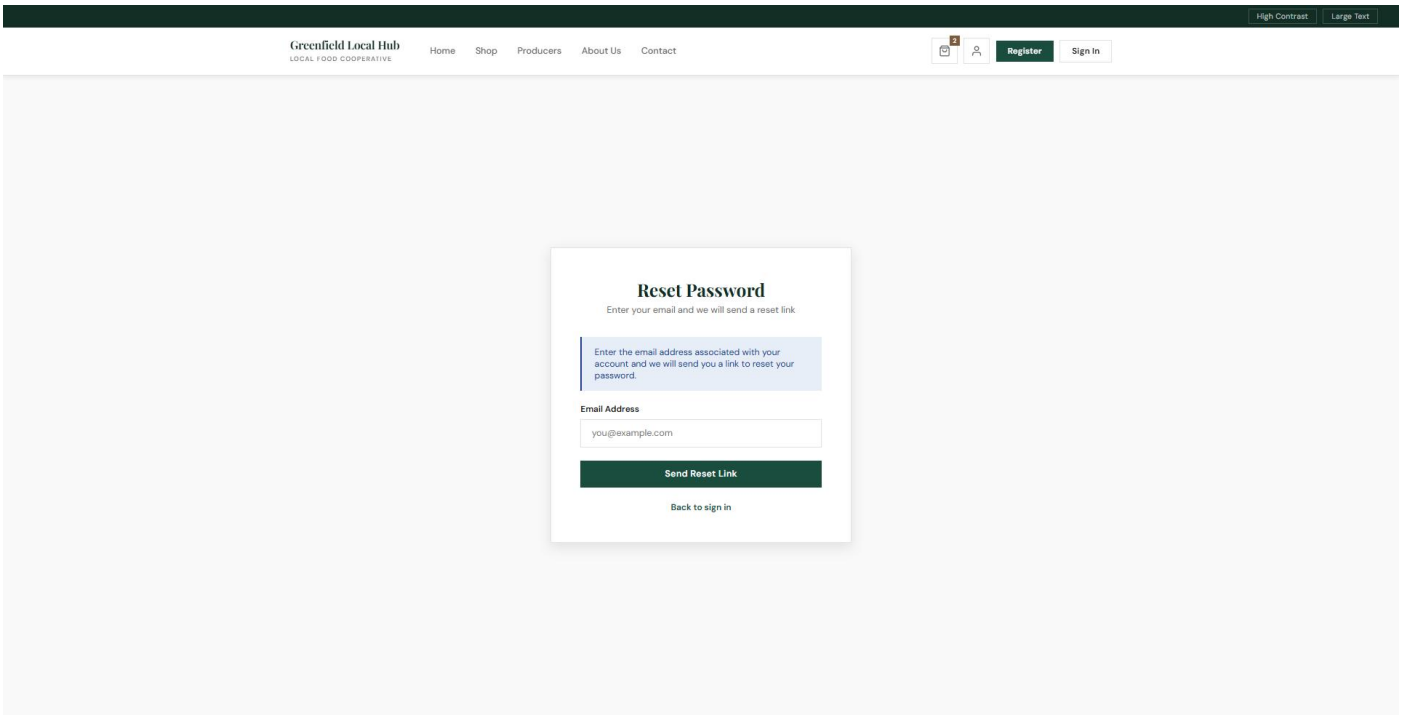
Sign in Page

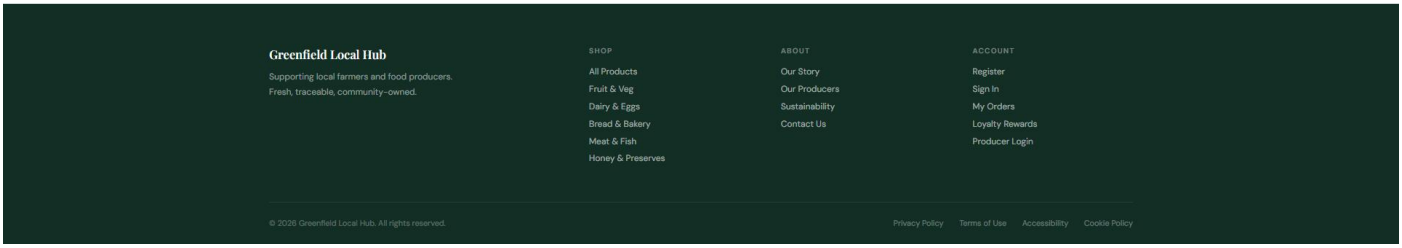




Similarly to the sign in page, this was designed to follow common convention. A simple but easy to complete sign in form with all necessary parts included.

Forgot Password Page

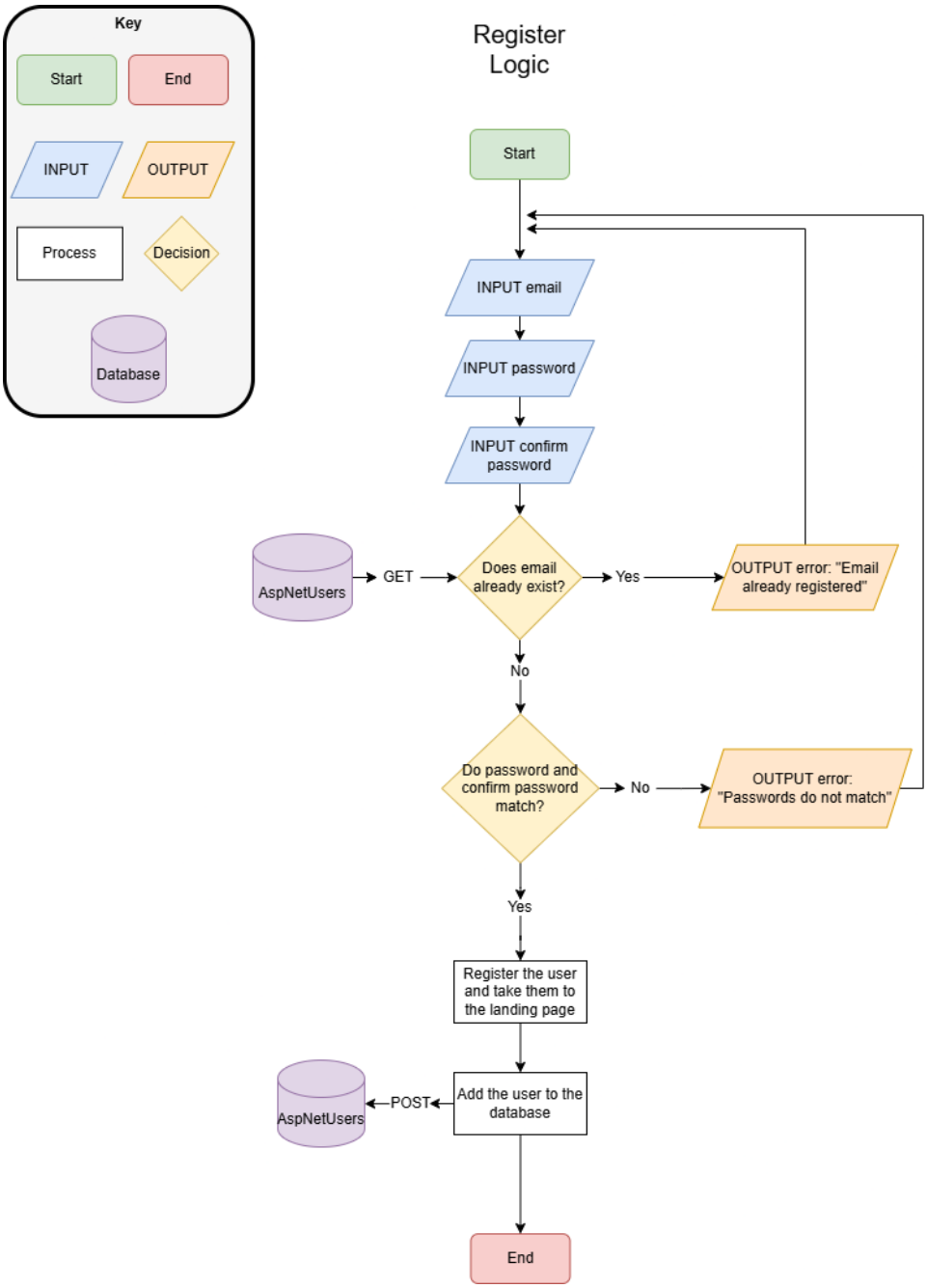




Lastly the forgot password page, including this page follows industry standard practice and is expected in all professional digital solutions. This small feature is important that greatly improves user experience by helping users regain access to their account. The user simply inputs their email address and clicks the button to receive an email with a reset link.

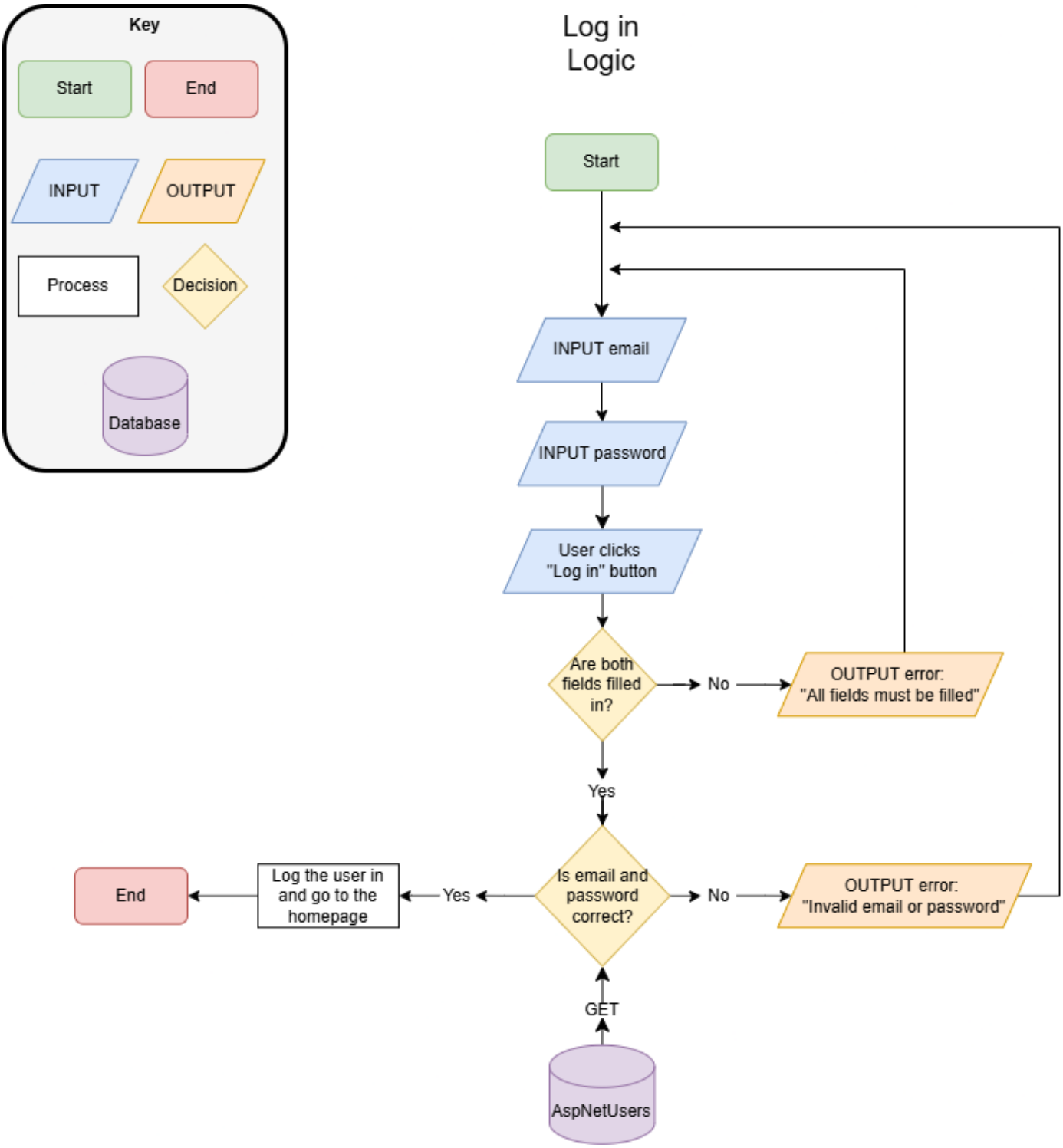
Algorithm Designs (Flowcharts)

Register Flowchart



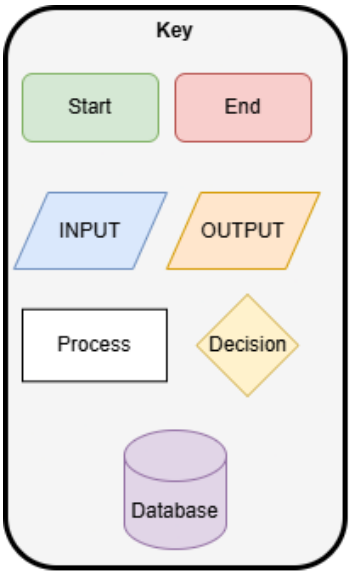
When a user wants to register an account, he will have to enter the email address, create a password and retype it to confirm it. The system will then check if the email has already been registered to another account before, if it is then an error message should appear. If the email has never been registered, the system will also check if the passwords match, if not an error message will also appear. If everything passes the validation checks, the user will then be registered and added to the database, and the solution will automatically redirect them to the homepage.

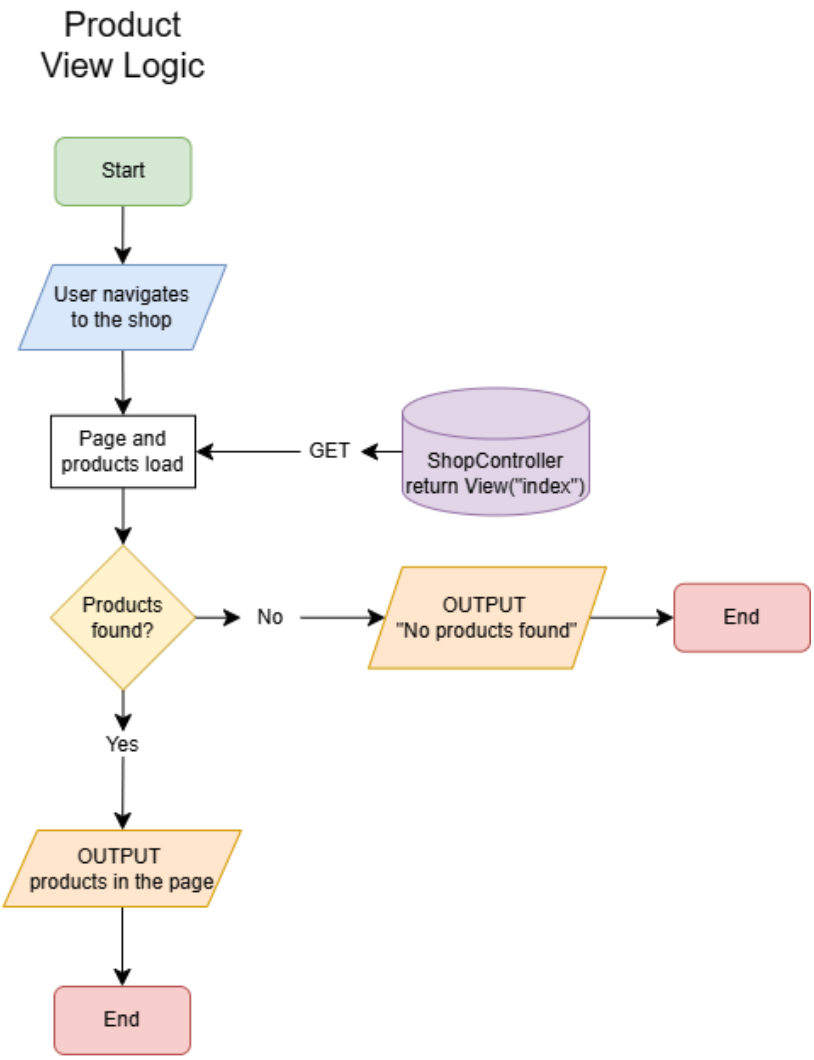
Sign in Flowchart



Once the user is in the log in page, they will have to enter their email address and password of the account in each of their respective fields. Now if the user clicks the sign in button after filling the required fields, the system will start to check if both fields are filled in. If not then an error message appears, if both fields are filled, the system will check if the email and password is a match that exists in the database. The database will use the get method to check the information, and if they do not match, then an error message appears for the user. If the log in credentials is valid then the user will be logged in and transferred to the homepage automatically.

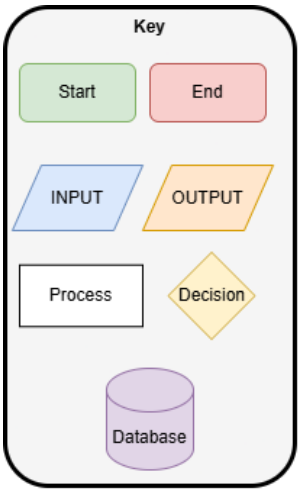
Product view Flowchart

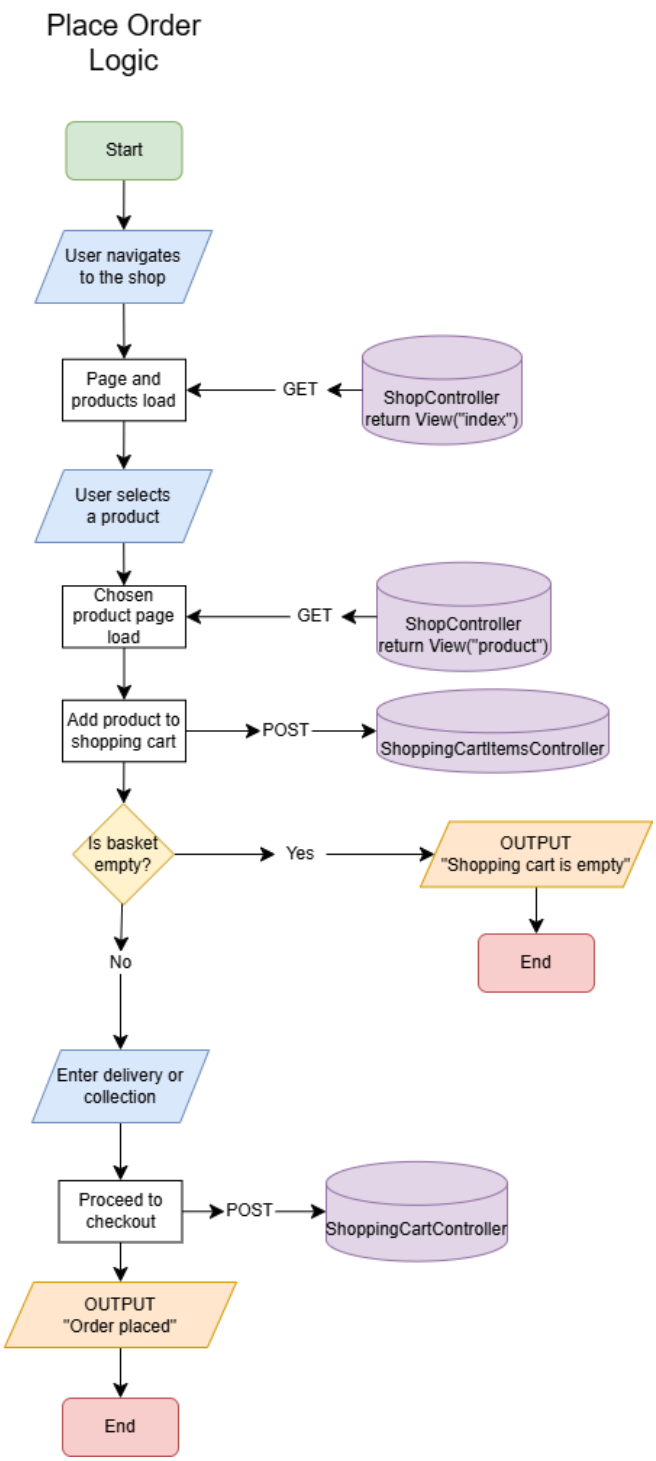




If a user wants to view all the products sold in the site, they will have to find the shop page using the navigation bar. After clicking the navigation link for the shop page, the page with all the content inside should load, and the shopController in the backend is responsible for that. If there are not any products in the page, a message will be visible for the user, but if there are products they will appear as intended.

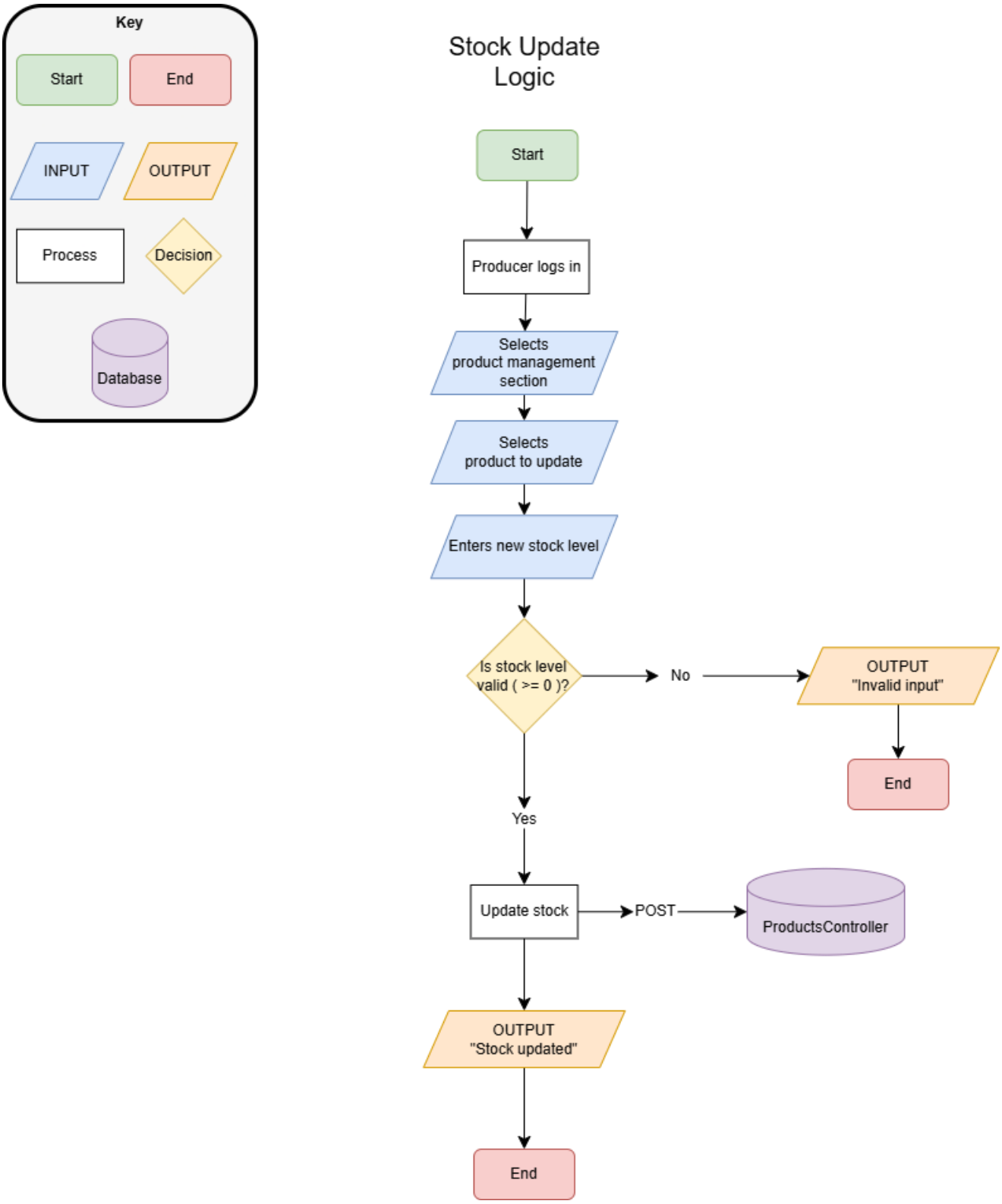
Place Order Flowchart





To place an order, the user must find the shop page first. The database will load the page using the get method, the user will then select the product he wants to buy after the page loaded. The specific product page will be displayed using the get method again, after that the user will have to add the product to the cart and this a post method in the backend since data will be added to the database this time. Now to make an order from the shopping cart it must check if it is empty first, if yes then a message will say that it is empty. If the cart is not empty, the user will have to choose if they want delivery or collection. When that has been done, the order will be added to the database using the post method. Finally, when that is successfully completed, a message stating that the order has been place will be displayed.

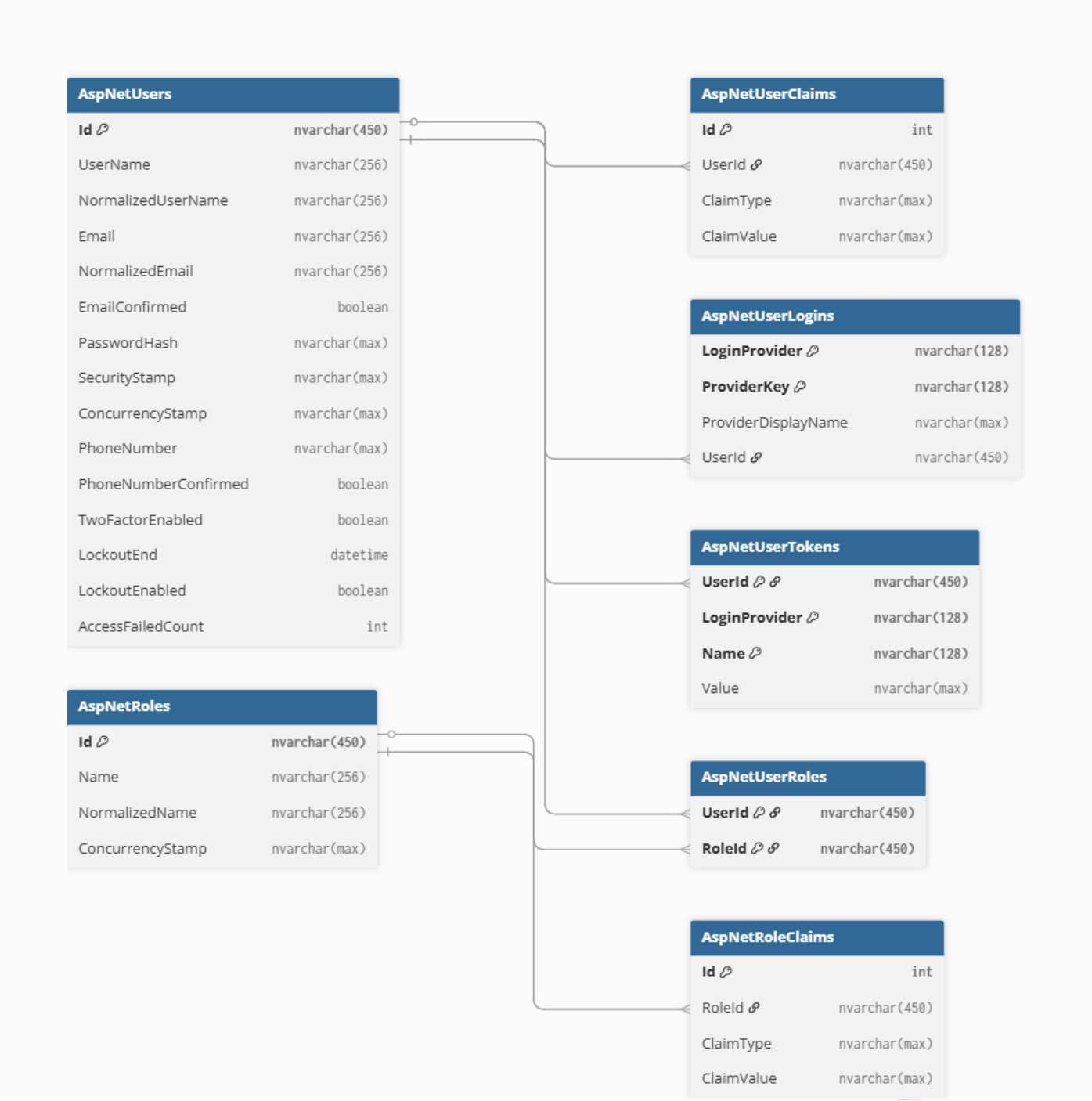
Stock Update Flowchart



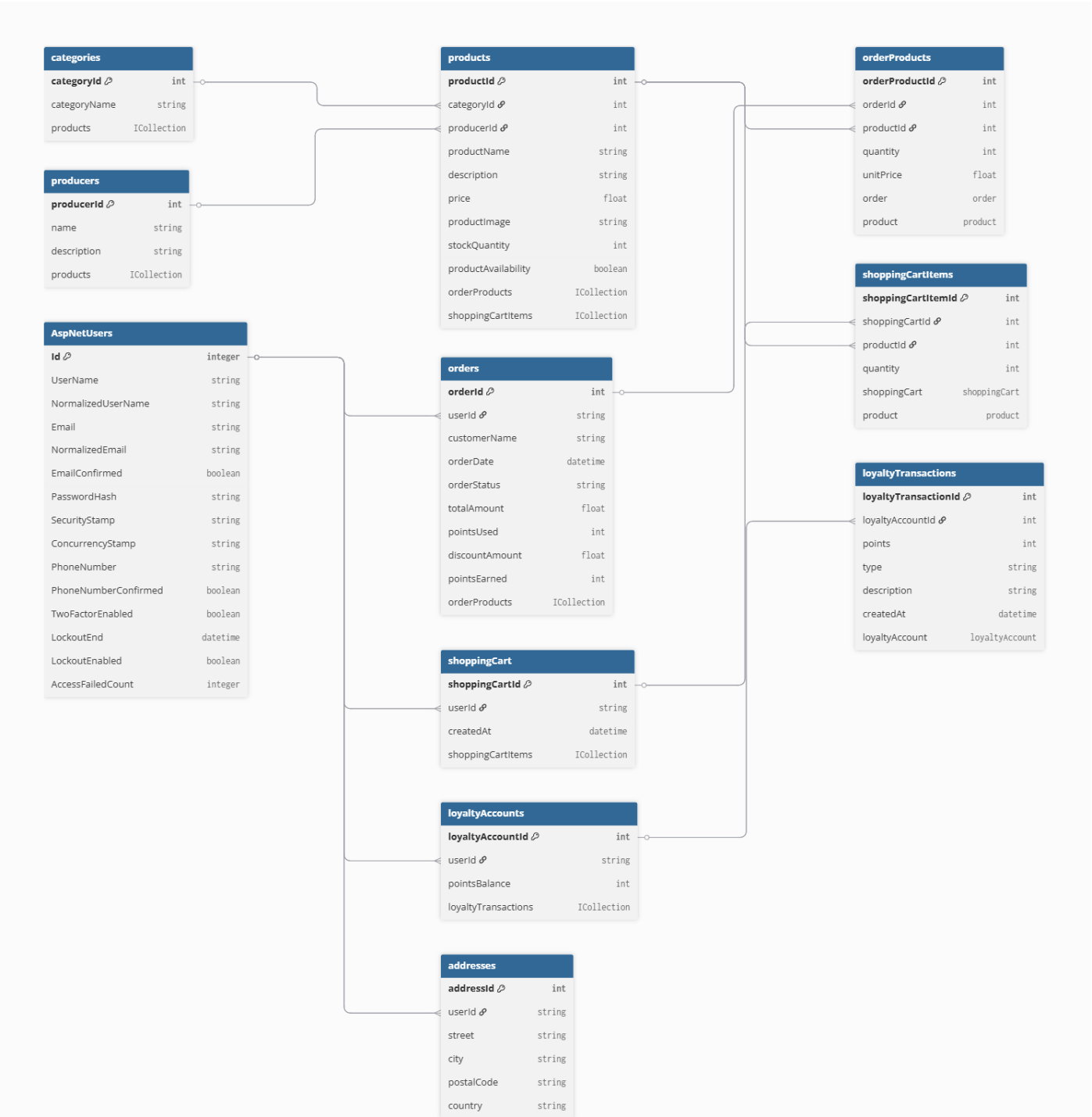
This flowchart shows how a producer would update his stock on a product. First, the producer must log in to his account, the producer would then navigate to the product management section in their producer dashboard. Inside the product management, the producer will find all the products that are posted and then they will select the product that requires updating. The producer will be able to enter the new stock level, the system will then validate if the value is zero or bigger. If it is not, then an error message will appear, but if it is a valid number then they can update the stock which will be posted to the database. After successfully updating, a message will display that the stock level is updated.

Database Diagram Design

MVC System Database Design



Greenfield Local Hub Database Design



Data Dictionary

MVC System Data Dictionary

AspNetUsers

Property	Datatype	Purpose	PK/FK	Size	Required?
Id	nvarchar	Primary identifier for the user	PK	450	Yes
UserName	nvarchar	The user's username	-	256	No
NormalizedUserName	nvarchar	Normalised username for lookups	-	256	No
Email	nvarchar	The user's email address	-	256	No
NormalizedEmail	nvarchar	Normalised email for lookups	-	256	No
EmailConfirmed	boolean	Shows if email is confirmed	-	Default	Yes
PasswordHash	nvarchar	Hashed password	-	MAX	No
SecurityStamp	nvarchar	Security stamp for password changes	-	MAX	No
ConcurrencyStamp	nvarchar	Checks if the user data has changed before saving.	-	MAX	No
PhoneNumber	nvarchar	User's phone number	-	MAX	No
PhoneNumberConfirmed	boolean	Shows if phone is confirmed	-	Default	Yes
TwoFactorEnabled	boolean	Shows if 2FA is enabled	-	Default	Yes
LockoutEnd	datetime	Date and time when lockout ends	-	Default	
LockoutEnabled	boolean	Shows if lockout is enabled	-	Default	Yes
AccessFailedCount	int	The amount of failed access attempts	-	Default	Yes

AspNetRoles

Property	Datatype	Purpose	PK/FK	Size	Required?
Id	nvarchar	Primary identifier for the role	PK	450	Yes
Name	nvarchar	The role name	-	256	No
NormalizedName	nvarchar	Normalised role name for lookups	-	256	No
ConcurrencyStamp	nvarchar	Checks if the role data has changed before saving	-	Max	No

AspNetUserRoles

Property	Datatype	Purpose	PK/FK	Size	Required?
UserId	nvarchar	Links the role task to a specific user	PK (Composite), FK	450	Yes
RoleId	nvarchar	Links the role task to a specific role	PK (Composite), FK	450	Yes

AspNetUserClaims

Property	Datatype	Purpose	PK/FK	Size	Required?
Id	int	A unique identifier for each user claim	PK	Default	Yes
UserId	nvarchar	Links claim to a specific user	FK	450	Yes
ClaimType	nvarchar	Describes what the claim is about	-	MAX	No
ClaimValue	nvarchar	Specifies the detail for that claim type	-	MAX	No

AspNetRoleClaims

Property	Datatype	Purpose	PK/FK	Size	Required?
Id	int	A unique identifier for each role claim	PK	Default	Yes
RoleId	nvarchar	Links claim to a specific role	FK	450	Yes
ClaimType	nvarchar	Describes what the claim is about	-	MAX	No
ClaimValue	nvarchar	Specifies the detail for that claim type	-	MAX	No

AspNetUserLogins

Property	Datatype	Purpose	PK/FK	Size	Required?
LoginProvider	nvarchar	Identifies the external login service like Google	PK (Composite)	128	Yes
ProviderKey	nvarchar	Stores the unique ID from the external provider for this user	PK (Composite)	128	Yes
ProviderDisplayName	nvarchar	Shows name for the login provider	-	MAX	No
UserId	nvarchar	Links the external login to a specific user	FK	450	Yes

AspNetUserTokens

Property	Datatype	Purpose	PK/FK	Size	Required?
UserId	nvarchar	Links the token to a specific user	PK (Composite), FK	450	Yes
LoginProvider	nvarchar	Identifies which provider the token belongs to	PK (Composite)	128	Yes
Name	nvarchar	The name of the token	PK (Composite)	128	Yes
Value	nvarchar	The value of the specified token	-	MAX	No

Greenfield Local Hub Data Dictionary

products

Property	Datatype	Purpose	PK/FK	Size	Required?
productId	int	A unique identifier for each product.	PK	Default	Yes
categoryId	int	Links a product to a category so users can filter the products in the shop page.	FK	Default	Yes
producerId	int	Links each product to a producer so the user can identify who is selling it.	FK	Default	Yes
productName	string	Name of the product displayed in the shop page.	-	Default	Yes
description	string	Extra details about the product.	-	Default	No
price	float	Value of the product	-	Default	Yes
productImage	string	Stores a specific image for each separate product	-	Default	No
stockQuantity	int	Amount of product available for sale.	-	Default	Yes
productAvailability	boolean	Displays if product is available or not.	-	Default	Yes

categories

Property	Datatype	Purpose	PK/FK	Size	Required?
categoryId	int	A unique identifier for each category.	PK	Default	Yes
categoryName	string	Stores the name of a category	-	Default	Yes

producers

Property	Datatype	Purpose	PK/FK	Size	Required?
producerId	int	A unique identifier for each producer.	PK	Default	Yes
name	string	Stores the name of a producer so it can be displayed in the products.	-	Default	Yes
description	string	Extra details about each producer.	-	Default	No

orders

Property	Datatype	Purpose	PK/FK	Size	Required?
orderId	int	A unique identifier for each order.	PK	Default	Yes
userId	string	Links the order to the user, so the user can track their orders only.	FK	Default	Yes
customerName	string	Stores the name used for the order.	-	Default	Yes
orderDate	datetime	Stores when the order was placed for history and tracking.	-	Default	Yes
orderStatus	string	Displays the state of the order.	-	Default	Yes
totalAmount	float	Stores the final price paid on the order.	-	Default	Yes
pointsUsed	int	Tracks how many points were used to reduce the price.	-	Default	No
discountAmount	float	Stores the amount reduced from the total price.	-	Default	No
pointsEarned	int	Stores the points gained from each order.	-	Default	No

orderProducts

Property	Datatype	Purpose	PK/FK	Size	Required?
orderProductId	int	A unique identifier for each product entry within an order.	PK	Default	Yes
orderId	int	Links the product to a specific order, so multiple products can exist in one order.	FK	Default	Yes
productId	int	Identifies which product was in the order.	FK	Default	Yes
quantity	int	Stores how many units of the product were ordered.	-	Default	Yes
unitPrice	float	Stores the value of the product at purchase time so the order history is accurate.	-	Default	Yes

shoppingCart

Property	Datatype	Purpose	PK/FK	Size	Required?
shoppingCartId	int	Identifies each shopping cart to a user session.	PK	Default	Yes
userId	string	Links the shopping cart to a specific user for a personalised shopping cart.	FK	Default	Yes
createdAt	datetime	Stores when the cart was created.	-	Default	Yes

shoppingCartItems

Property	Datatype	Purpose	PK/FK	Size	Required?
shoppingCartItemId	int	A unique identifier for each product in the shopping cart.	PK	Default	Yes
shoppingCartId	int	Links the item to a specific cart so multiple products can belong to one cart.	FK	Default	Yes
productId	int	Identifies which product is added to the cart.	FK	Default	Yes
quantity	int	Stores the amount of the product the user wants to buy.	-	Default	Yes

loyaltyAccounts

Property	Datatype	Purpose	PK/FK	Size	Required?
loyaltyAccountId	int	A unique identifier for each users loyalty points tracker.	PK	Default	Yes
userId	string	Links the loyalty account to a specific user.	FK	Default	Yes
pointsBalance	int	Stores the amount of points available for the user to use.	-	Default	Yes

loyaltyTransactions

Property	Datatype	Purpose	PK/FK	Size	Required?
loyaltyTransactionId	int	A unique identifier for each points transaction.	PK	Default	Yes
loyaltyAccountId	int	Links the transaction to a loyalty account to update points balance.	FK	Default	Yes
points	int	Stores how many points were added or removed.	-	Default	Yes
type	string	Shows if points were earned or spent to define the action.	-	Default	Yes
description	string	Extra details about the transaction.	-	Default	No
createdAt	datetime	Stores when the transaction happened.	-	Default	Yes

Addresses

Property	Datatype	Purpose	PK/FK	Size	Required?
addressId	int	A unique identifier for each address so a user can store multiple addresses.	PK	Default	Yes
userId	string	Links the address to a user.	FK	Default	Yes
street	string	Stores the street/road name and house number for delivery.	-	Default	Yes
city	string	Stores the city for delivery location.	-	Default	Yes
postalCode	string	Stores the postal code to help identify the exact area.	-	Default	Yes
country	string	Stores the country.	-	Default	Yes